

Comparing GDP Nowcasting Models in the United States and Türkiye

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Abstract

Official GDP statistics are released with substantial delays, creating an informational gap for policymakers and market participants who must assess economic conditions in real time. This paper conducts a pseudo-real-time comparison of seventeen GDP nowcasting models (spanning classical time-series benchmarks, penalized linear regressions, tree-based machine learning, neural networks, mixed-frequency regressions, and Bayesian/factor methods) across two contrasting economic environments: the United States, a data-rich advanced economy, and Türkiye, a shorter-history, more volatile emerging market. Information availability is simulated through publication-lag and ragged-edge masking at three standardized within-quarter vintages, while GDP targets use final revised data. The evaluation covers 2017-2025 for the United States (36 quarters) and 2018-2025 for Türkiye (32 quarters). Our central finding is that model rankings are not portable across countries. In the United States, flexible high-dimensional methods perform best: the LSTM roughly halves the RMSFE of the ARMA benchmark at the late within-quarter vintage, followed by the MLP and a reduced-dimensional Bayesian VAR. In Türkiye, shrinkage and parsimonious econometric models dominate: Lasso, ElasticNet, and BVAR lead the full-sample ranking, while the LSTM falls behind the univariate benchmark. These advantages are concentrated in turbulent periods: the leading models earn most of their edge during the COVID window, whereas, in calm quarters, the differences between models shrink, and the univariate benchmark is hard to beat. Accuracy also improves substantially as within-quarter information accumulates, with the largest gains going to each country’s best models. The results indicate that data richness, sample length, and macroeconomic volatility shape the relative usefulness of econometric and machine-learning nowcasting methods, cautioning against one-size-fits-all model choices in real-time policy analysis.

Keywords: Nowcasting; GDP; Real-time forecasting; Mixed-frequency econometrics; Dynamic factor models; Machine learning; Neural networks; Cross-country comparison; Emerging markets; Türkiye.

Code availability: [GitHub](#)¹

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¹https://github.com/omerfyldz/nowcasting_gdp_turkey_us

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1 Introduction

National income accounts, and Gross Domestic Product (GDP) in particular, have served as standard indicators of economic performance since their development in the 1930s (Kuznets, 1934; Landefeld et al., 2008). Yet despite their central role in economic analysis and policymaking, official GDP statistics are released with significant delays. First estimates appear roughly at $t + 30$ days in the United States, and $t + 60$ in Türkiye (Bureau of Economic Analysis, nd; Presidency of Strategy and Budget, 2026). This release lag creates a recurring informational gap for policymakers, financial institutions, and market participants who must assess economic conditions in real time.

The cost of this gap is far from theoretical. Central banks adjust their policy instruments at a much higher frequency than GDP is published, and financial markets continuously revise expectations about growth and embed them into asset prices. The need for timely estimates becomes especially important during fast-moving episodes such as crises: during the 2008 Global Financial Crisis, the long delays in the release of macroeconomic data made it difficult for policymakers to direct counter-cyclical measures (Cascaldi-Garcia et al., 2023). These considerations gave rise to *nowcasting*, a term that refers to the prediction of the very recent past, the present, and the very near future. Originally used in meteorology, it was introduced into economics by Giannone et al. (2008). The literature has since broadened to include mixed-frequency methods such as MIDAS regressions and mixed-frequency VAR models (Forni and Marcellino, 2014) and, more recently, a wide range of machine-learning techniques (Richardson et al., 2021).

The relative performance of these alternative approaches, however, is far from settled. Predictive accuracy varies substantially with the economic environment, the richness of high-frequency data, and the structural characteristics of the economy under study. Most existing studies focus on a single country or region (predominantly the United States), and the available evidence is still insufficient to draw clear conclusions about how alternative model classes compare across heterogeneous macroeconomic environments (Stundziene et al., 2024).

This paper contributes to closing that gap. We conduct a unified empirical comparison of seventeen GDP nowcasting models across two contrasting economic contexts: the United States, a data-rich advanced economy, and Türkiye, a data-limited emerging-market economy with high macroeconomic volatility. The seventeen models span one univariate benchmark (ARMA) and six methodological families: classical econometric models (OLS, VAR); penalized linear regressions (Ridge, Lasso, ElasticNet); tree-based machine learning (Decision Tree, Random Forest, Gradient Boosting, XGBoost); neural networks (Multilayer Perceptron, LSTM, DeepVAR); mixed-frequency methods (MIDAS, MIDAS-ML); and Bayesian/factor-based methods (BVAR, Dynamic Factor Model).

The empirical design is related to the benchmark-comparison tradition in GDP nowcasting, including Hopp (2022), which evaluates multiple nowcasting methodologies under a common real-time framework and provides an open-source implementation. Building on this tradition, we adapt and substantially modify parts of that implementation to fit the objectives of this study: comparing seventeen specifications, rebuilding the data pipeline for both the United States and Türkiye, incorporating country-specific predictor panels and release calendars, and updating the evaluation

window through 2025-Q4. These changes allow us to compare not only individual models, but also the robustness of model-family rankings across different macroeconomic environments and across the COVID and post-COVID periods.

Three aspects of the empirical design are worth highlighting. First, the exercise is *pseudo-real-time*: information availability is simulated through publication-lag and ragged-edge masking, but GDP targets use final revised data rather than historical first-release vintages. Second, each model produces nowcasts at three standardized within-quarter information sets ($\mathbf{m1}$, $\mathbf{m2}$, $\mathbf{m3}$), so that the value of within-quarter information accumulation can itself be measured. Third, the model lineup is identical across the two countries while the predictor panels are country-specific, which lets us ask not whether a single model “wins,” but whether the *ranking* of model families is stable across very different data environments.

Our empirical results show that it is not. In the United States, flexible and high-dimensional methods lead the full-sample ranking at the most informative within-quarter vintage: the LSTM attains a root mean squared forecast error (RMSFE) about 50% of the ARMA benchmark, followed by the MLP and a reduced-dimensional Bayesian VAR. In Türkiye, the same lineup produces nearly the opposite ordering: Lasso, ElasticNet, and BVAR lead, parsimonious OLS and VAR specifications remain competitive, and the LSTM falls behind the univariate ARMA benchmark. This cross-country contrast is not only a full-sample result: it is strongest in the full sample and during the COVID stress test. In the calmer quarters the models perform much more alike and the advantage over the simple ARMA benchmark shrinks sharply, in Türkiye’s post-COVID quarters ARMA is in fact the most accurate model of all. Accuracy improves substantially from the early to the late within-quarter vintages, and the biggest improvements go to each country’s top models; the post-release horizons confirm the main rankings and also show the validation-selected Turkish dynamic factor model becoming highly competitive once one or two months of post-quarter data arrive.

The remainder of the paper is organized as follows. Section 2 reviews the nowcasting literature and positions our contribution. Section 3 describes the construction of the two country panels, the stationarity transformations, the variable-selection procedure, and the vintage conventions. Section 4 presents the seventeen models and the evaluation protocol, including the mathematical structure of the main model classes. Section 5 reports the empirical results. Section 6 interprets the cross-country differences and relates them to the literature. Section 7 concludes with limitations and directions for future research.

2 Literature Review

2.1 Development of the Nowcasting Literature

Modern economic nowcasting is commonly associated with [Giannone et al. \(2008\)](#), who formalized the problem of updating current-quarter GDP forecasts as macroeconomic releases arrive in real time and demonstrated the framework using United States GDP. Their framework asks how an analyst should exploit the staggered, real-time arrival of macroeconomic indicators to produce a continuously updated estimate of quarterly GDP before the official release becomes available. The conceptual basis had already been developed by [Mariano and Murasawa \(2003\)](#), who showed

how monthly and quarterly series could be combined within a single coincident business-cycle index. This provided an early mixed-frequency state-space framework that later nowcasting studies built on.

Bańbura et al. (2013) provide the standard survey of the field’s consolidation, organizing the literature around three structural challenges that any candidate nowcasting model must confront: the *mixed-frequency* nature of the data, in which the target is observed less often than its predictors; the *ragged-edge* pattern of release lags, which means that the available panel has different end dates for different series at any given moment; and the *high dimensionality* of the predictor space relative to the number of training observations, which increases the risk of overfitting in traditional regression estimators. These three challenges motivate every methodological choice in the present study. Bok et al. (2018), describing the New York Fed Staff Nowcast, provide a practical reference point for US GDP nowcasting with a broad macroeconomic information set and scheduled data releases; their data-flow logic informs the construction of our m1/m2/m3 vintages.

2.2 Methodological Strands

Four broad methodological strands define the contemporary nowcasting toolkit, and the seventeen models compared in this paper span all four.

Dynamic factor models. The DFM-based nowcasting tradition associated with Giannone et al. (2008) remains central to central-bank practice. A small number of latent common factors are assumed to summarize the co-movement of a large panel of indicators, and GDP growth is then forecasted using these estimated factors. In state-space implementations, the Kalman filter allows the model to update the factors even when the data arrive at different frequencies and with different publication lags. Stock and Watson (2002) establish the foundations for estimating large factor models by principal components, Doz et al. (2012) provide the quasi-maximum-likelihood theory, and Bańbura and Modugno (2014) extend DFM estimation to incomplete datasets by using an expectation–maximization procedure, which iteratively estimates the unobserved factors and updates the model parameters. This is important for nowcasting because real-time indicators are released with different lags and frequencies, and thus the panel data is rarely complete.

Mixed-frequency regressions. The MIDAS framework of Ghysels et al. (2004) approaches the frequency mismatch from a different angle: high-frequency predictors enter the regression at their own frequency, while parsimonious lag-weighting schemes limit the number of parameters. Clements and Galvão (2008) show that MIDAS regressions can effectively exploit monthly indicators for forecasting US output growth, and provide support for combining forecasts produced by separate single-indicator MIDAS models. Foroni et al. (2015) propose unrestricted MIDAS, a less constrained version of MIDAS that is useful when the frequency mismatch is small, such as between monthly indicators and quarterly GDP, and Kuzin et al. (2011) compare MIDAS with mixed-frequency VARs for the euro area. One of the recent extensions is the MIDAS-ML framework of Babii et al. (2022), which combines mixed-frequency regression with high-dimensional regularization. The method uses basis functions to approximate MIDAS lag

weights and estimates the model with sparse-group lasso, making MIDAS more suitable for large predictor sets.

Bayesian VARs. Bayesian vector autoregressions address dimensionality through informative shrinkage rather than dimension reduction. In large macroeconomic panels, the Minnesota-type prior used by [Bańbura et al. \(2010\)](#) shrinks the coefficient matrix toward a parsimonious benchmark, combining the stability of a univariate model with a multivariate information set. This makes BVARs particularly useful in nowcasting settings where the number of potential predictors is large relative to the number of quarterly GDP observations. [Hopp \(2022\)](#) reports the Bayesian VAR as one of the two best-performing methodologies in his US GDP nowcasting benchmark, alongside the LSTM.

Machine learning. The most recent strand applies machine-learning algorithms to macroeconomic forecasting and nowcasting. Penalized regression entered the macroeconomic forecasting toolkit early, with [De Mol et al. \(2008\)](#) showing that shrinkage methods such as lasso and ridge can perform similarly to principal-components factor models when applied to large macroeconomic panels. More flexible nonlinear methods have since been used to capture interactions and regime-dependent relationships that are difficult to impose in standard linear specifications. [Goulet Coulombe et al. \(2022\)](#) show that nonlinear machine-learning methods are especially useful under conditions such as high macroeconomic uncertainty and financial stress. [Richardson et al. \(2021\)](#) find that machine-learning algorithms outperform simple benchmarks for New Zealand GDP in a real-time nowcasting setting, and [Soybilgen and Yazgan \(2021\)](#) combine dynamic factors with tree-based ensembles for US GDP nowcasting. For neural networks, [Hopp \(2022\)](#) finds the LSTM architecture of [Hochreiter and Schmidhuber \(1997\)](#) to be the best overall performer in a twelve-model US GDP nowcasting benchmark, with especially strong performance during the COVID period.

2.3 Comparative Benchmark Studies

A growing body of work takes the side-by-side comparison of competing nowcasting methods as the central object of study. [Hopp \(2022\)](#) is the closest precursor to this paper: he compares twelve methodologies for US GDP nowcasting using the [Bok et al. \(2018\)](#) predictor set, three crisis episodes, and five simulated data vintages, finding LSTM and Bayesian VAR to be the strongest overall performers. [Kant et al. \(2025\)](#) reach a different conclusion for Dutch GDP, where the random forest provides the most accurate forecasts and nowcasts. [Dauphin et al. \(2022\)](#) apply DFMs and several machine-learning algorithms to six European economies and report that machine-learning methods reduced average forecast errors by up to seventy-five percent relative to an AR(1) benchmark during the pandemic period. Most recently, [Akepanidtaworn and Akepanidtaworn \(2025\)](#) evaluate a broad set of traditional econometric and machine-learning models in Monte Carlo simulations and six country case studies. They reach a more skeptical conclusion: traditional econometric methods are the best performers in four of six countries, while simpler machine-learning methods such as lasso and elastic net generally outperform more complex nonlinear models.

The disagreement among these benchmarks is itself informative. The US evidence in [Hopp \(2022\)](#) favors LSTM and Bayesian VAR; [Kant et al. \(2025\)](#) favor random forest for Dutch GDP; [Dauphin et al. \(2022\)](#) report strong machine-learning gains during the pandemic; and [Akepanidtaworn and Akepanidtaworn \(2025\)](#) find stronger support for traditional econometric methods and simple linear machine-learning models. This suggests that nowcasting performance is highly context-dependent, varying with the country, sample length, data richness, and evaluation period. The present paper is designed to test this sensitivity directly by holding the model lineup fixed while comparing a data-rich advanced economy with a shorter and more volatile emerging-market dataset.

2.4 Nowcasting GDP in Türkiye

A relatively small literature focuses specifically on Turkish GDP nowcasting, but covers several important nowcasting approaches. Two features distinguish the Turkish case from the standard US setting. First, Turkish GDP is released with a substantially longer delay, roughly two months after the end of the reference quarter, and no advance estimate is produced ([Modugno et al., 2016](#)). Second, the macroeconomic data infrastructure is much younger, so training samples are shorter and more vulnerable to structural breaks such as the 2001 banking crisis, the 2008 global financial crisis, the 2018 currency crisis, and the COVID-19 shock.

[Akkoyun and Günay \(2012\)](#) adapt the Stock-Watson coincident-indicator model to Turkish GDP with a limited mixed-frequency dataset and establish that even a small model can outperform a univariate benchmark. [Modugno et al. \(2016\)](#) deploy a fifteen-variable dynamic factor model and show that it produces lower mean squared errors than the institutional nowcasts of the Central Bank of the Republic of Türkiye, the IMF, and the OECD over 2008–2013; their news-decomposition analysis attributes nowcast revisions to individual data releases. [Soybilgen and Yazgan \(2017\)](#) revisit this analysis after the substantial 2016 revisions to the Turkish GDP series, finding that essentially all relevant information for nowcasting Turkish GDP arrives within the contemporaneous quarter and that credit variables improve DFM performance while broad financial-market variables deteriorate it. [Günay \(2020\)](#) studies Turkish GDP nowcasting with MIDAS regressions and shows that forecast performance depends on the lag-polynomial specification, lag length, forecast horizon, and choice of indicator. He finds that no single specification dominates across all cases, while unrestricted MIDAS with a small number of lags can be competitive once within-quarter monthly data become available.

2.5 Position of the Present Study

The present paper sits at the intersection of these strands. With respect to the benchmark-comparison tradition, it is closest to [Hopp \(2022\)](#), but differs by comparing seventeen specifications, adapting and substantially modifying the empirical pipeline for a two-country setting, and extending the evaluation window through 2025-Q4 so that the post-COVID recovery becomes observable alongside the shock itself. With respect to the Turkish literature, it compares a broad set of model families for Turkish GDP under the same evaluation framework, building on the DFM work of [Modugno et al. \(2016\)](#) and [Soybilgen and Yazgan \(2017\)](#) and the MIDAS-based work of [Günay \(2020\)](#). With respect to the broader multi-country benchmark literature of

Dauphin et al. (2022) and Akepanidtaworn and Akepanidtaworn (2025), it contributes a country pair that differs substantially in sample length, data structure, and macroeconomic volatility. This design allows us to examine whether the relative performance of nowcasting models changes across country environments, as suggested by the literature’s finding that no single model is universally superior across datasets, countries, and time periods.

3 Data

This study uses two parallel datasets covering, respectively, a data-rich advanced economy (the United States) and a data-limited emerging-market economy (Türkiye). The two datasets share a common target, sample architecture, and pseudo-real-time evaluation; while differing in source, dimensionality, and time coverage.

3.1 Target Variable and Sample Design

The target variable in both countries is the quarterly log change in seasonally adjusted real GDP, placed at the end-of-quarter month. For the United States we use FRED’s chained-2017-dollar real GDP series GDPC1 from the FRED-QD database (McCracken and Ng, 2021); for Türkiye we use the corresponding seasonally adjusted real GDP series NGDPRSAXDCTRQ. In both cases the target is

$$y_t = \log(\text{GDP}_t) - \log(\text{GDP}_{t-3}), \quad (1)$$

where t is in months and y_t takes a non-missing value only when t is the closing month of a quarter (March (03), June (06), September (09), or December (12)). This log-difference is the standard transformation for making the GDP series stationary in the FRED-MD/FRED-QD framework (McCracken and Ng, 2016), and measures real GDP growth over the quarter. We report this raw quarterly growth rather than an annualized rate; as every model is scored against the same target.

Predictions are evaluated against the latest-available revised value of the target, as recorded in the 2026-03 FRED-QD vintage. Forecast errors are therefore computed against the revised GDP value in the final dataset, not the contemporaneous first release. We do not attempt a full real-time-vintage evaluation. Such an exercise would require reconstructing, for each forecast origin, not only the GDP release available at that date but also the corresponding historical vintages of all monthly predictors, their publication lags, and subsequent revisions. While this is partly feasible for the United States through archival sources such as ALFRED, comparable long-span, machine-readable vintage archives are substantially harder to obtain for Türkiye. Our exercise is therefore best interpreted as a *pseudo-real-time* evaluation: the information set is restricted within the quarter, but forecast errors are computed relative to a fixed, latest-available target vintage. This convention follows the simulated-vintage logic of Hopp (2022), while differing from the full real-time-vintage backtesting framework of Bok et al. (2018), which evaluates nowcasts using reconstructed historical data vintages.

The sample is split chronologically into training, validation, and out-of-sample test windows, with the validation window used for hyperparameter and specification selection (Table 1).² For

²All models share the same validation and test windows. The training window also coincides across models

the United States, the validation window (2008-2016) deliberately contains the Global Financial Crisis, so hyperparameters are stress-tested on a major shock before the COVID test period. Türkiye’s split is directed mainly by its shorter sample; its 2012-2017 validation window covers the 2013 Taper Tantrum but no comparable recession, and the 2018 currency crisis falls in the test window (again, due to the shorter sample size).

Table 1: Sample partitioning, by country.

	United States	Türkiye
Training window	1959-Q1 to 2007-Q4	1995-Q1 to 2011-Q4
Validation window	2008-Q1 to 2016-Q4	2012-Q1 to 2017-Q4
Test window	2017-Q1 to 2025-Q4	2018-Q1 to 2025-Q4
Test quarters	36	32

To absorb the pandemic periods’ shocks, three quarter-specific dummy variables (2020-Q2, 2020-Q3, and 2020-Q4) are added to the predictor set, and are used in all models. The post-COVID time-series literature emphasizes that pandemic observations should not be treated as ordinary data points, since they can strongly distort parameter estimates (Lenza and Primiceri, 2022). We therefore include COVID-quarter dummies as a simple adjustment. As these are calendar-based indicators, they use ex-post knowledge and should be interpreted as a modeling adjustment rather than strictly real-time information.

3.2 United States Panel

The US predictor pool combines three sources: (i) the FRED-MD database of monthly macroeconomic series (McCracken and Ng, 2016); (ii) the quarterly series in FRED-QD that are not already present in FRED-MD (McCracken and Ng, 2021) (variables placed on the monthly grid at quarter-end months); and (iii) a supplementary set of roughly twenty high-relevance indicators collected manually (including the Chicago Fed National Activity Index, the National Financial Conditions Index, ISM purchasing managers’ indices, regional Fed manufacturing surveys, and the S&P 500). After removing duplicates and stationarity transformations (Section 3.4), the pool contains 296 transformed variables on a monthly grid, plus the three COVID dummies. All series are drawn from the same release vintage (2026-03) for internal consistency.

Four series are additionally used at weekly frequency, exclusively for the mixed-frequency models (MIDAS and MIDAS-ML): initial claims for unemployment insurance (ICSA), the weekly Chicago Fed NFCI (NFCI), and a trade-weighted dollar index (constructed from DTWEXM and DTWEXBGS, where the latter is used after the former is discontinued). Daily values are aggregated to a weekly grid. These weekly series are assigned zero publication lag: they are truncated at the forecast vintage date but are not lagged further back, because they are released within a few days.

except for the mixed-frequency models (MIDAS and MIDAS-ML), whose start dates are set by the availability of the weekly indicators they require: 1967-Q1 for the United States and 2002-Q1 for Türkiye.

3.3 Türkiye Panel

The Turkish predictor pool is constructed from scratch, since no curated Turkish counterpart to FRED-MD exists. Three sources are combined: (i) the Electronic Data Delivery System (EVDS) of the Central Bank of the Republic of Türkiye (CBRT), which contributes the majority of macro-financial indicators (industrial production, consumer and producer prices, broad money, the real effective exchange rate, the USD/TRY exchange rate, the financial account, reserves, confidence indicators, capacity utilization, and credit and deposit measures); (ii) the Turkish Statistical Institute (TÜİK), which contributes labour-market and real-activity indicators (employment and unemployment, tourism arrivals, vehicle production, and turnover-index series); and (iii) Investing.com for the BIST 100 equity index. After removing duplicates, and the stationarity transformations below, the pool contains 54 transformed monthly predictors plus the GDP target and the three COVID dummies. Two series are used at native weekly frequency for the mixed-frequency models: the weekly consumer loan interest rate and the weekly deposit interest rate, both from CBRT.

A central complication of the Turkish data is that indicators begin in very different years, so their histories within the training window differ widely. Variables are therefore split by data coverage: the 22 series with sufficient history ($\approx 30\%$ non-missing in the training window) form the main panel used by all models, while 32 short-history series are placed in a separate “Tier-C” group reserved for the dynamic factor model, which can in principle absorb sparse series through its latent-factor structure. As discussed in Section 4.3, however, these Tier-C series were ultimately excluded as the estimation routine (`nowcastDFM`) failed on the resulting sparse panel.

3.4 Stationarity Transformations

Each raw series is assigned a transformation code (tcode) chosen to make it stationary, using the seven-code system of [McCracken and Ng \(2016\)](#). For the United States, these codes are taken directly from the FRED-MD and FRED-QD release files, which supply a recommended transformation for every series. We change the default for only two series: `NWPIx`, which trends strongly over time, and `HWIx`, whose definition changed during the sample. Both are reset to the log first difference.

For Türkiye, transformations applied manually in two steps. First, twenty non-seasonally-adjusted real-economy series are deseasonalized using STL decomposition. Second, each variable’s tcode is chosen by a step-by-step procedure that allows for a structural break in the series. We first apply the Zivot-Andrews unit-root test ([Zivot and Andrews, 1992](#)) to the level of each series. We start with this test because standard unit-root tests can wrongly conclude that a series is non-stationary when it actually contains a structural break ([Perron, 1989](#)), a real concern for Türkiye, where the sample includes the 2001 and 2018 crises. If this test does not find the series stationary, we turn to the standard Augmented Dickey-Fuller test and apply it to a sequence of transformed versions of the series, stopping at the first version that tests stationary. We begin with the series in levels: if it is already stationary, it needs no transformation (tcode 1). If not, we test its first difference; if that is stationary, the series enters the models in first differences (tcode 2). If it is still not stationary, we test the log first difference and use that form (tcode 5). A series that passes none of these tests is assigned the log first difference by default. The

resulting distribution over the 55 series (54 predictors plus the target) is 7 in levels, 24 in first differences, and 24 in log first differences.

All transformed series are then re-checked with a battery of standard stationarity tests, and in neither panel is any series classified non-stationary (confirming that the chosen transformations remove the unit roots). For the United States, we use three tests; Augmented Dickey-Fuller (ADF), KPSS, and Phillips-Perron (PP), and classify a series as stationary only when all three agree. The 296 predictors split into 220 stationary and 76 inconclusive. For Türkiye we add a fourth, structural-break-aware test, Zivot-Andrews, and classify a series as stationary when at least three of the four tests support it; the 55 series split into 48 stationary and 7 inconclusive. The inconclusive cases are kept in the set as the majority of tests still support stationarity in each one: in the US panel they typically arise when KPSS flags a mild structural shift while ADF and PP continue to support stationarity, whereas the Turkish borderline cases are split two-to-two across the four tests.

3.5 Variable Selection and Predictor Categories

The US pool of 296 transformed predictors was far larger than unpenalized multivariate models could accommodate properly. We therefore narrow the candidate pool through a four-method ensemble feature-selection procedure, run once on the training window of each country and held fixed before any model estimation begins: a cross-validated Lasso, a cross-validated ElasticNet, a tuned random forest paired with permutation importance, and Lasso stability selection (Meinshausen and Bühlmann, 2010) based on 100 subsamples drawn without replacement at 75% of the training rows. All cross-validation respects the time ordering of the data and is carried out only on the pre-test training windows (1959-2007 for the United States, 1995-2011 for Türkiye). Standardization is applied inside each cross-validation fold, so that information from the hold-out fold never leaks into the scaling step. We deliberately include a nonlinear method (random forest) alongside the linear ones, since disagreement between linear ones may indicate predictive content is beyond simple linear correlation. This is especially clear in the United States, where Lasso and the random forest share only 5 of their top 35 variables.

The selection results map onto a three-category that determines each model’s information set:

- **Category 1** contains only the target and serves the univariate ARMA benchmark.
- **Category 2** serves the unpenalized multivariate models (OLS and VAR), which can only use a small number of predictors. For the United States it contains four real-activity variables (`outbs`, `outnfb`, `gcec1`, `houstne`) (the top picks from the random forest and from stability selection) and for Türkiye it contains `ipi_sa`, `usd_try_avg`, `cpi_sa`, and `fin_acc`. Together with the three COVID dummies, each Category-2 set has seven features.
- **Category 3** serves the remaining models, which can handle many predictors because they rely on regularization, ensembling, or shrinkage. For the United States it gathers every variable chosen by at least one of the four selection methods at its strict cutoff, giving 53 variables (56 features with the COVID dummies); for Türkiye it is simply the 22 predictors with enough data coverage (25 features with the dummies).

Penalized and machine-learning models handle the full Category-3 panel internally through shrinkage, regularization, or ensembling; the BVAR and DFM instead use further reduced or validation-selected sets for computational feasibility, as documented in Section 4.

3.6 Forecast Vintages and Pseudo-Real-Time Conventions

For each test quarter Q with closing month T , nowcasts are produced at standardized information dates (Table 2). All vintages target the same quarter- Q realization; they differ only in the information available when the forecast is made. The **m1/m2/m3** triple is the symmetric cross-country benchmark. In addition, *post-release* robustness horizons are produced (**post1** for both countries and **post2** for Türkiye) in which one or two months of post-quarter data are available while the GDP target quarter itself is still unpublished. Because Turkish GDP is released roughly two months after quarter-end (versus about a month in the United States), these post-release horizons are operationally meaningful for Türkiye; they answer a different question than the within-quarter vintages and are reported separately.

Table 2: Forecast vintages. T denotes the closing month of the target quarter.

Vintage	Information date	Interpretation
m1	$T - 2$	earliest; closest to a one-quarter-ahead forecast
m2	$T - 1$	mid-quarter
m3	T	the nowcast with the most within-quarter information
post1	$T + 1$	robustness: post-quarter information, same target
post2	$T + 2$	robustness, Türkiye only

In real-time nowcasting, indicators arrive with different publication lags, so the available data have a “ragged edge” (Bańbura and Modugno, 2014). To reproduce this, each variable is given a publication-lag value in a metadata table. For the United States we set these lags series by series from each source’s release calendar, following Bok et al. (2018), taking one of three values. The quarterly National Income and Product Accounts (NIPA) series are given a two-month lag (this group includes the GDP target and its components). This is a deliberately cautious choice that keeps the current quarter’s figures out of the early vintages, even though the first official GDP estimate itself appears about one month after the quarter ends. The most timely series (financial indicators, surveys, and the headline employment and industrial-production releases) are given a zero lag because they come out early in the following month, and the remaining monthly indicators, released around mid-month, are given a one-month lag. For Türkiye the same source-based logic is applied to the CBRT and TÜİK release calendars, with the GDP target carrying the longest lag.

At each forecast date, the information set is constructed by applying the relevant publication lag to each variable, so that observations that would not yet have been released are excluded from the model’s available data. What changes across vintages is only the forecast date, not the lag rule. The GDP target row for the current quarter is kept in the panel so that mixed-frequency models can produce a nowcast, but its value is hidden from every model before prediction. Weekly indicators are assigned zero publication lag: they are truncated at the forecast vintage date but

are not lagged further back, because they are released within a few days. No publication lag is imposed on the COVID dummies, which are fixed by construction.

One mechanical implication is worth noting. Here the two-month figure refers to the publication-lag value assigned to the GDP target in the metadata mask, which is distinct from the institutional release delay discussed above (about a month in the United States and two months in Türkiye). Because this masking value is two months, the univariate ARMA benchmark has the same effective GDP history at `m1` and `m2`, and its predictions at these two vintages coincide by construction; multivariate models still differ across all vintages as their indicator sets update monthly.

4 Methodology

4.1 Forecasting Framework

We estimate all 17 models using a shared pseudo-real-time forecasting design. We make minor adjustments only when an estimator requires model-specific steps. The forecast vintages match the three standardized within-quarter information sets from Section 3.6. Table 1 reports the country-specific sample partitions. For models with hyperparameters, we select values using time-ordered cross-validation. These values remain fixed throughout the test period (see Table 13 in Appendix A). We strictly avoid standard K-fold cross-validation. That approach would incorrectly mix future and past observations within a fold. During the test period, we re-estimate most models at every forecast date. We use only the data available up to that specific milestone. The estimation window therefore grows as the exercise moves forward. This approach follows the standard expanding-window design in the nowcasting literature (Giannone et al., 2008; Bok et al., 2018). The main exception is the dynamic factor model. We estimate the DFM once on the training window and run it forward using the Kalman filter (Section 4.3).

The flattened supervised models share a common preprocessing sequence at each iteration. First, we exclude observations based on their publication lag. This ensures the model only sees information available at that exact forecast date (Section 3.6). Second, we fill any remaining missing values with the column mean. We calculate this mean using only the training portion of that specific iteration. This mean-filling approach is computationally simple. It performs similarly to EM or ARMA methods at a fraction of the computational cost in similar benchmarks (Hopp, 2022). Third, we scale predictors when necessary using parameters from the current training slice. Penalized regressions use standard scaling with zero mean and unit variance. The MLP uses a robust scaler. We leave COVID dummies unscaled. Fourth, the model fits the data and generates a nowcast. The system writes this nowcast to disk, ensuring full reproducibility. To capture high-frequency movements, these models expand each monthly predictor. The pipeline includes the current value plus three monthly lags. We then drop all months except quarter-ends. This layout allows the regression to run at a quarterly frequency while still capturing within-quarter monthly dynamics.

The state-space, mixed-frequency, and recurrent neural network models bypass this flattened pipeline. The DFM handles ragged-edge data through its state-space representation and the Kalman filter. We estimate the BVAR inside the mixed-frequency state-space setup of our

implementation. MIDAS and MIDAS-ML keep predictors at their original frequencies. They do not flatten the panel into a quarterly matrix first. LSTM and DeepVAR rely on their own internal sequence construction and scaling. Some models depend on random initialization or stochastic estimation. For these architectures, we lock in fixed seeds and combine the forecasts at each date. We average the tree ensembles, LSTM, and DeepVAR across ten seeded runs. For the MLP, we use the median prediction across ten seeded runs. We do not average the single decision tree.

We perform feature selection before the out-of-sample test period begins. This step uses only pre-test data: pre-2008 for the United States and pre-2012 for Türkiye. This partition keeps variable selection separate from test-period outcomes. It prevents our model comparison from using ex-post information about the evaluation sample. Preprocessing order is critical here. If test-period information entered feature selection, imputation, scaling, or hyperparameter choices, the out-of-sample results would be invalid. We ran a leakage audit after completing the pipeline. The audit confirmed that the saved files contain no missing or non-finite values. It also verified that targets match perfectly across models and that no tuning code relies on test-period outcomes.

4.2 Benchmark Time-Series Models

ARMA The univariate benchmark is an ARMA model on the target history alone,

$$y_t = c + \sum_{i=1}^p \phi_i y_{t-i} + \sum_{j=1}^q \theta_j \varepsilon_{t-j} + \varepsilon_t \quad (2)$$

with the order selected by the Akaike information criterion on the training window (ARMA(2,0) in both countries) and the model re-estimated at every forecast origin on the publication-lag-masked GDP history. All relative accuracy measures and Diebold-Mariano tests in Section 5 are computed against ARMA in the same country, panel, and vintage; the implementation and interpretation of the Diebold-Mariano tests are described in Section 4.8.

OLS The OLS benchmark regresses the target on the Category-2 predictor set: each of the four indicators enters at its contemporaneous value and three monthly lags, alongside the three COVID dummies. Writing $x_{j,t-\ell}$ for Category-2 indicator j observed ℓ months before the quarter-end month t , and $D_{k,t}$ for the COVID dummies,

$$y_t = \beta_0 + \sum_{j=1}^4 \sum_{\ell=0}^3 \beta_{j,\ell} x_{j,t-\ell} + \sum_{k=1}^3 \gamma_k D_{k,t} + \varepsilon_t \quad (3)$$

estimated by ordinary least squares (with an intercept) on the quarter-end months at every forecast origin. The variables used here are not chosen manually, but come from the Category-2 feature-selection rule described in Section 3; in the United States this rule selects two output-related measures, a government-spending measure, and a regional housing-starts series, so the OLS and VAR benchmarks should be read as data-driven small-information benchmarks rather than structural models.

Unlike ARMA, this OLS specification contains no lagged dependent variable: rather than adding indicators on top of the autoregressive structure, it replaces the target’s own-lag dynamics with contemporaneous and lagged indicator information. The ARMA-to-OLS comparison therefore reflects the combined effect of removing the univariate autoregressive dynamics and introducing a small cross-sectional information set. The VAR in equation (4) re-introduces the lagged target that OLS omits, so the ARMA $\rightarrow$ OLS $\rightarrow$ VAR sequence is not a strictly nested ladder.

VAR The VAR benchmark estimates a vector autoregression

$$\mathbf{z}_t = \mathbf{c} + A_1 \mathbf{z}_{t-1} + \mathbf{u}_t \quad (4)$$

on the quarterly system $\mathbf{z}_t$, which stacks the target together with the lag-expanded Category-2 indicators. The endogenous lag order is fixed at one, so each variable depends on the previous quarter’s values, and any column that does not vary in the training window (mainly the COVID dummies, which are zero before 2020) is dropped, since a zero-variance column would make the system singular. Relative to OLS, the VAR adds dynamic feedback between the target and the predictors through the autoregressive system.

4.3 Bayesian and Factor-Based Methods

BVAR The Bayesian VAR maintains the vector autoregression structure of (4) but applies it to a larger, mixed-frequency dataset. To handle the high number of resulting coefficients, it applies a Minnesota-type prior (Bańbura et al., 2010). This prior shrinks the coefficient matrix toward a simple autoregressive benchmark instead of estimating an unrestricted high-dimensional VAR. Because our target variables are growth rates rather than levels, we interpret this prior as shrinkage toward weak own-lag persistence. This differs from a standard random-walk prior that assumes unit own-lag coefficients. A single tightness parameter, λ , controls the overall strength of this shrinkage. Smaller values impose heavier regularization. We fix $\lambda = 0.1$ for both countries, following a common default in the large-BVAR literature. We do not tune this parameter via cross-validation. This choice creates an intentional asymmetry compared to our penalized regressions. Repeated cross-validation over λ is computationally prohibitive in a mixed-frequency state-space setup. Furthermore, we use the BVAR as a disciplined baseline rather than a fully optimized machine learning model. We estimate the model using the `mbvar` R package. This package manages the mix of monthly and quarterly data, alongside ragged edges, within a state-space framework.

Two distinct computational limits shape this implementation. First, the full Category-3 indicator system was too large for `mbvar` to estimate successfully. For the United States, we use a smaller predictor set selected via Lasso on the pre-2008 training window to prevent data leakage. This subset includes the variables that account for the first 80% of cumulative Lasso importance. This step yields 15 predictors plus the GDP target, which we label the “Lasso-80” set. For Türkiye, we use the locked Category-2 indicators along with the GDP target. Second, during the 2025-Q4 `m3` and `post1` vintages, the reduced US covariance matrix becomes numerically singular. For those two specific horizons, we fall back to forecasts generated by the smaller Category-2

system. Consequently, all BVAR results in this study stem from reduced-size mixed-frequency variants rather than full, large-scale systems. We flag the two US fallback cases in the evaluation summary in Section 4.8.

DFM The dynamic factor model assumes that a large panel of indicators $\mathbf{x}_t$ is driven by a few unobserved common factors,

$$\mathbf{x}_t = \Lambda \mathbf{f}_t + \mathbf{e}_t, \quad \mathbf{f}_t = A_1 \mathbf{f}_{t-1} + \cdots + A_p \mathbf{f}_{t-p} + \mathbf{u}_t \quad (5)$$

where $\mathbf{f}_t$ are the common factors, Λ maps them to the observed series, and $\mathbf{e}_t$ are the series-specific (idiosyncratic) errors, which may be serially correlated. The model is written in state-space form and estimated with the EM algorithm of Bańbura and Modugno (2014), while the Kalman filter deals with the missing values and hence the ragged edge directly. We use the `nowcastDFM` R package released with Hopp (2022), and group the variables into the blocks of Bok et al. (2018): one global factor plus separate factors for thematic groups such as surveys, real activity, and the labour market. Unlike the expanding-window models, the DFM parameters are estimated once and then kept fixed; each nowcast is produced by running the Kalman filter forward through the publication-lag-adjusted data for that vintage, the way DFMs are usually used in practice.

The US DFM combines monthly Category-3 indicators with the quarterly GDP target. This specification includes 27 monthly indicators organized into the block structure described above, plus the GDP target, totaling 28 variables. We exclude other quarterly indicators and COVID dummies for practical reasons. The software package maps each quarterly series to five additional states, meaning too many variables quickly exhaust system memory. Furthermore, the COVID dummies lack variation before 2020, which breaks the data standardization step. We estimate the US DFM on the pre-test sample ending in 2016-Q4. We then filter it forward through the 2017–2025 test period. This setup lacks COVID-quarter controls, and we do not re-estimate its parameters during the test period. Therefore, its pandemic-era forecasts reflect this specific implementation rather than the performance of factor models in general.

The Turkish DFM required a separate variable selection step. Estimating on the full sparse panel, which includes 32 short-history Tier-C variables, caused numerical failures inside `nowcastDFM`. To resolve this, we compared three candidate predictor sets: Category 2, a 10-variable set, and Category 3. We evaluated them using their 2012–2017 validation RMSFE, averaged across the `m1`, `m2`, and `m3` vintages. Category 2 performed best and was selected. We then retrained the model through 2017 before starting the 2018–2025 test period (see Figure 10 in Appendix A). This choice relied entirely on the validation window rather than test data. Thus, our final Turkish DFM is a feasible, validation-selected factor model instead of a full-panel specification. We estimate the Turkish DFM on the sample ending in 2017-Q4 and filter it forward through the 2018–2025 test period.

4.4 Mixed-Frequency Econometric Approaches

MIDAS MIDAS regressions (Ghysels et al., 2004) link the quarterly target directly to higher-frequency predictors, keeping each predictor at its own frequency. For a single indicator x

observed m times per quarter, the regression is

$$y_t = \alpha + \beta \sum_{j=0}^{J-1} w_j(\boldsymbol{\theta}) x_{t-j/m} + \varepsilon_t \quad (6)$$

where the lag weights $w_j(\boldsymbol{\theta})$ follow an exponential Almon function, so that just a few parameters $\boldsymbol{\theta}$ describe a smooth pattern of weights across all the lags. This keeps the parameter count small, since giving every high-frequency lag its own free coefficient would require far too many parameters. Following the forecast-combination idea of [Clements and Galvão \(2008\)](#), we fit one MIDAS regression per indicator (monthly indicators with $J = 4$ lags and $m = 3$; weekly indicators with $J = 13$ and $m = 13$) using the `midasr` R package, and then combine the resulting single-indicator nowcasts in a weighted average based on each indicator model’s in-sample fitted RMSE.

Since this is a cross-country comparison, we define the weighting rules for both nations explicitly. For the United States, weights are proportional to the inverse in-sample RMSE of each single-indicator MIDAS model. We normalize these values to sum to one. Consequently, better-fitting indicators receive higher weights. However, the worst-fitting indicator still retains a small, positive weight. For Türkiye, weights are proportional to the distance between each indicator’s in-sample RMSE and the largest in-sample RMSE in the pool. We normalize these values to sum to one as well. Under this specific rule, the worst-fitting indicator receives a weight of zero. Both rules utilize in-sample fitted RMSE instead of an independent validation RMSE. This difference in weighting methods is an implementation artifact rather than a deliberate design choice. The Turkish MIDAS results should be interpreted with this constraint in mind.

The US combination leverages the Category-3 predictor set. We enter each variable at its observed frequency whenever feasible. We also incorporate the four weekly indicators. The Turkish combination uses the Category-3 monthly predictors alongside the two weekly interest rate series. We exclude the monthly versions of these two series to prevent double counting.

MIDAS-ML The machine-learning version of MIDAS, due to [Babii et al. \(2022\)](#), replaces the per-indicator combination with one large joint regression. Each indicator’s high-frequency lags are summarized by a small set of Legendre polynomial terms, and all indicators are then estimated together under a sparse-group lasso (sg-LASSO) penalty,

$$\hat{\boldsymbol{\beta}} = \arg \min_{\boldsymbol{\beta}} \left\{ \frac{1}{2n} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda \left[\gamma \|\boldsymbol{\beta}\|_1 + (1 - \gamma) \sum_{g \in \mathcal{G}} \|\boldsymbol{\beta}_g\|_2 \right] \right\} \quad (7)$$

where each group g holds the basis terms of one indicator. The penalty has two parts: the ℓ_1 part shrinks individual coefficients toward zero, while the group part can switch off a whole indicator at once. The parameter γ sets the balance between the two parts, and λ sets the overall strength. We use a degree-3 Legendre basis and set $\gamma = 0.25$, within the framework of [Babii et al. \(2022\)](#), and estimate the model with the `midasm1` R package. For the United States, λ is chosen by cross-validation on the training window (`cv.sglmfit`). For Türkiye, this cross-validation step repeatedly crashed the notebook process, so the completed Turkish results instead use a fixed penalty, `sglmfit`($\lambda = 0.01$). The Turkish MIDAS-ML results should therefore

be read as a constrained implementation of the method, not as evidence against MIDAS-ML in general.

4.5 Penalized Linear Regression

After expanding each predictor with monthly lags, the penalized-regression design matrices are high-dimensional relative to the available quarterly sample. The US Category-3 design matrix has about 200 columns for roughly 200 quarterly observations in the pre-test sample. The Turkish design matrix is smaller, with about 90 columns, but the available estimation sample is also much shorter: roughly 67 quarters in the initial training window and about 90 quarters when the validation window is included. Thus, in both countries the number of regressors is large relative to the number of observations, and in Türkiye the design is close to the boundary where ordinary least squares becomes unstable. The three penalized regressions replace OLS with a slightly biased but much more stable estimator through the elastic-net objective

$$\hat{\beta} = \arg \min_{\beta} \left\{ \frac{1}{2n} \|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \alpha \left[\rho \|\beta\|_1 + \frac{1-\rho}{2} \|\beta\|_2^2 \right] \right\} \quad (8)$$

which gives the Lasso of Tibshirani (1996) at $\rho = 1$, the Ridge of Hoerl and Kennard (1970) at $\rho = 0$, and the ElasticNet of Zou and Hastie (2005) in between. The ℓ_1 part sets some coefficients exactly to zero, so it also performs variable selection, while the ℓ_2 part shrinks all coefficients smoothly and spreads the weight across correlated predictors. Here α controls the overall strength of the penalty, and ρ controls the balance between the two parts.

Penalty strengths are selected using time-ordered cross-validation on the training and validation portions of the sample and are then kept fixed while the model is refit at every forecast date. The selected specifications indicate a clear cross-country difference in the amount of shrinkage required. In the United States, the selected penalties are very small, and the ElasticNet specification effectively behaves like Lasso. In Türkiye, the selected penalties imply substantially stronger shrinkage, with the ElasticNet leaning much more heavily toward the Ridge component. This pattern is consistent with the shorter and noisier Turkish sample, where stronger regularization is needed to stabilize the fitted relationship.

4.6 Tree-Based Machine Learning

Tree-based methods split the predictors into rectangular regions and fit a constant value in each region, which lets them capture nonlinear effects and interactions without the modeller specifying them in advance. We include four members of this family: a single decision tree; the Random Forest of Breiman (2001), which averages many trees, each grown on a random sample of the data and a random subset of the predictors; gradient boosting (Friedman, 2001), which builds the trees one after another,

$$F_M(\mathbf{x}) = \sum_{m=1}^M \nu h_m(\mathbf{x}), \quad h_m \approx -\partial L / \partial F_{m-1} \quad (9)$$

where each new shallow tree h_m is fitted to the portion of the target that the current ensemble still mispredicts. This process targets the negative gradient of the squared-error loss. The model

then incorporates each new tree with a small learning rate, ν . Similarly, XGBoost (Chen and Guestrin, 2016) is a refined boosting variant. It adds an ℓ_2 penalty to the leaf values and utilizes a second-order Taylor approximation of the loss function.

All four methods use the flattened Category-3 predictors without standardization. This step is unnecessary because tree splits are invariant to feature scale. We select their hyperparameters via time-ordered cross-validation and keep them fixed throughout the test period. For the three ensemble methods (Random Forest, Gradient Boosting, and XGBoost), we average predictions across ten random seeds at each forecast date. In contrast, we fit the single decision tree only once using its optimal hyperparameters.

The cross-validated parameters for Türkiye are generally more restricted than those for the United States. This difference echoes the heavy regularization pattern seen in our penalized regressions. However, the models enforce this restriction through different levers. Türkiye relies on smaller ensembles for Gradient Boosting and XGBoost. It also requires larger minimum leaf sizes for the Random Forest, along with shallower trees for both the single decision tree and XGBoost.

One structural trait of these methods heavily influences our empirical results. Because tree-based models fit a constant value within each leaf region, they cannot extrapolate outside the range of target values seen in the training data. As Hopp (2022) demonstrates, this bounded prediction window creates severe issues during extreme anomalies like the 2020-Q2 nowcasts.

4.7 Neural Network Models

MLP The multilayer perceptron (MLP) passes the flattened predictor matrix through fully connected hidden layers with ReLU activations to predict the target. We train the model using the Adam optimizer. Since MLPs are highly sensitive to outliers and feature scales, this implementation applies a robust scaling step instead of the standard scaling used for the penalized linear models. Grid search determines the network size and the strength of the ℓ_2 weight-decay penalty. For both countries, validation selects a small network with two hidden layers (32 and 16 units) and strong weight decay. We also incorporate early stopping with an internal validation split and a patience rule to limit overfitting during training. At each forecast date, we train ten networks from different random initializations. We report the median prediction across these ten runs rather than the mean. The Turkish MLP remains subject to the early-vintage instability caveat discussed below.

LSTM The long short-term memory (LSTM) network (Hochreiter and Schmidhuber, 1997) uses gated memory cells to bypass the vanishing-gradient problem. This challenge typically prevents ordinary recurrent networks from learning from long sequences. Gated cells allow the model to process monthly indicators as a time-ordered sequence instead of a flattened cross-section. We implement the `nowcast_lstm` library from Hopp (2022). Built specifically for mixed-frequency datasets, this package ingests a monthly panel alongside a quarterly target while managing data scaling and ragged edges internally. The network utilizes a six-month lookback window. Cross-validation selects a small architecture consisting of a single layer with 32 hidden units. At each forecast date, we train ten networks with distinct random starts and average their

predictions.

DeepVAR DeepVAR (Salinas et al., 2019) extends recurrent neural network forecasting to a fully multivariate setting. A stacked recurrent network forecasts all variables jointly, and we extract the target nowcast from this combined output. The model builds closely on the autoregressive recurrent architecture of Salinas et al. (2020). However, the multivariate DeepVAR framework remains the primary reference for our joint-output implementation. The original GluonTS/MXNet code from Hopp (2022) is no longer maintained. Instead, we deploy a PyTorch version that preserves the core architecture: four stacked LSTM layers, 40 hidden units, and a 12-month context window. We apply identical settings in both countries to ensure comparability. This PyTorch implementation relies on its own internal scaling and sequence construction rather than the flattened pipeline used by our other supervised models.

One important caveat is entirely mechanical. With a 12-month context window, transitioning from m1 to m3 alters only the tail end of the input sequence. Missing-value imputation then smooths much of this structural variation away. Consequently, DeepVAR nowcasts show minimal variation across the three within-quarter vintages in both countries. We treat this lack of vintage sensitivity as a known limitation when analyzing performance. Specifically, any apparent post-COVID accuracy from DeepVAR reflects this flat-forecast artifact rather than pure predictive skill.

4.8 Evaluation Protocol

For every model, country, vintage, and evaluation panel, forecasts are scored with the Root Mean Squared Forecast Error (RMSFE) and the Mean Absolute Error (MAE),

$$\text{RMSFE} = \sqrt{\frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} (\hat{y}_t - y_t)^2}, \quad \text{MAE} = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} |\hat{y}_t - y_t|. \quad (10)$$

Here y_t is the final revised quarterly log GDP change of (1) and $\mathcal{T}$ is the set of test quarters in the panel. Losses are computed in log-difference units without annualization. For comparability across countries and panels we also report the relative RMSFE against the univariate benchmark,

$$\text{relRMSFE}_{\text{model}} = \frac{\text{RMSFE}_{\text{model}}}{\text{RMSFE}_{\text{ARMA}}}. \quad (11)$$

This ratio is computed within the same country, panel, and vintage. Values below one indicate improvement over ARMA. Statistical comparisons against ARMA use the Diebold-Mariano test (Diebold and Mariano, 1995) on the squared-error loss differential $d_t = e_{\text{model},t}^2 - e_{\text{ARMA},t}^2$. Because the evaluation panels are short, ranging from 7 to 36 quarters, and the implementation uses a simple loss-differential variance, DM results are reported as directional evidence and interpreted cautiously.

To prevent a single full-sample summary from masking regime-dependent performance, results are reported over four country-specific panels (Table 3). The quarter 2020-Q1 is part of the full panel but not of any sub-period panel, because the sub-periods treat 2020-Q2 as the start of the COVID window. For Türkiye we label the early panel “pre-crisis” rather than “pre-COVID”,

because it already contains the 2018 currency crisis and so is not a calm period in the way the US pre-COVID panel is. Indeed, this difference between the two countries is itself informative.

Table 3: Evaluation panels, by country.

Panel	United States		Türkiye	
	Range	Quarters	Range	Quarters
Pre-COVID / pre-crisis	2017-Q1–2019-Q4	12	2018-Q1–2019-Q4	8
COVID	2020-Q2–2021-Q4	7	2020-Q2–2021-Q4	7
Post-COVID	2022-Q1–2025-Q4	16	2022-Q1–2025-Q4	16
Full	2017-Q1–2025-Q4	36	2018-Q1–2025-Q4	32

4.9 Implementation Caveats

For transparency, Table 4 collects the implementation constraints documented above. None of these is hidden in the results: where a constrained implementation underperforms, we attribute the result to the implementation rather than to the model class.

Table 4: Implementation caveats to be kept in mind when reading the results.

Item	Caveat
Pseudo-real-time	Information availability is simulated through publication-lag rules and ragged-edge construction; GDP targets are final revised values; revision uncertainty is not measured.
Coarse vintages	Three standardized within-quarter information sets are evaluated, not the full release-by-release data flow; no news decomposition is performed.
BVAR (both countries)	Reduced-dimensional: US uses a Lasso-80 predictor set, with documented Category-2 fallbacks at the 2025-Q4 <code>m3</code> and <code>post1</code> vintages because of a singular covariance matrix; Türkiye uses locked Category-2 predictors plus the target.
US DFM	Estimated once on the pre-test sample and filtered forward; quarterly indicators and COVID dummies are excluded for implementation reasons, so pandemic-period errors should be read as implementation-specific.
Türkiye DFM	Validation-selected Category-2 monthly panel plus target, chosen on 2012-2017 validation RMSFE and retrained through 2017; Tier-C short-history variables are excluded after estimation failed on the sparse panel.
MIDAS	Single-indicator nowcasts are combined using in-sample fitted RMSE weights, but the exact weighting rule differs across countries; the Türkiye MIDAS result should be read with this implementation artifact in mind.
Türkiye MIDAS-ML	Fixed-penalty <code>sglfit</code> with $\lambda = 0.01$ after rolling cross-validation aborted the computation; a constrained implementation.
Türkiye MLP	Early-vintage (<code>m1/m2</code>) outputs are finite but unstable; read as model instability, not as a headline result.
DeepVAR (both countries)	Predictions are nearly identical across <code>m1/m2/m3</code> because tail information changes are smoothed by imputation.
DM tests	Simple loss-differential variance and small panels; interpret significance cautiously.
<code>post1/post2</code>	Robustness horizons answering a different question; not replacements for the symmetric <code>m1/m2/m3</code> benchmark.

5 Empirical Results

This section presents our main empirical results. All metrics are computed from saved prediction files via our country-aware evaluation script. They are fully reproducible using the repository files.³ We organize our analysis around five key questions. First, which models win at the main nowcast vintage (§5.1)? Second, how does performance vary across sub-periods, particularly during COVID-19 (§5.2)? Third, how much does within-quarter information accumulation matter (§5.3)? Fourth, do the rankings hold up during post-release robustness horizons (§5.4)? Finally, what value do forecast combinations add in the United States (§5.5)?

5.1 Full-Sample Rankings at the m3 Vintage

Tables 5 and 6 show the full 17-model rankings on the complete test panel at the m3 vintage. This vintage contains the maximum amount of within-quarter information. Crucially, these full-sample rankings are heavily *crisis-weighted*. As shown in Section 5.2, the extreme volatility of the COVID-19 quarters dominates the pooled RMSFE. Consequently, this ranking reflects model performance under severe stress rather than a stable, all-weather ordering.

Table 5: United States: 17-model ranking on the full panel (2017-Q1 to 2025-Q4) at the m3 vintage. Relative RMSFE measures performance against the ARMA benchmark. Two-sided Diebold-Mariano (DM) p -values evaluate squared-error loss relative to ARMA. For models with a relative RMSFE above one, a low p -value means the model is significantly *worse* than ARMA.

Rank	Model	RMSFE	MAE	Rel. RMSFE	DM p -value
1	LSTM	0.0116	0.0062	0.502	0.151
2	MLP	0.0137	0.0065	0.592	0.144
3	BVAR	0.0146	0.0102	0.633	0.240
4	Ridge	0.0152	0.0070	0.660	0.150
5	MIDAS	0.0182	0.0076	0.791	0.236
6	MIDAS-ML	0.0186	0.0076	0.807	0.336
7	ElasticNet	0.0189	0.0078	0.818	0.236
8	Lasso	0.0189	0.0078	0.818	0.236
9	Gradient Boosting	0.0192	0.0078	0.831	0.318
10	Decision Tree	0.0193	0.0080	0.836	0.346
11	XGBoost	0.0194	0.0080	0.841	0.355
12	OLS	0.0194	0.0080	0.842	0.355
13	DeepVAR	0.0195	0.0080	0.845	0.356
14	Random Forest	0.0195	0.0080	0.845	0.366
15	VAR	0.0198	0.0082	0.861	0.357
16	ARMA	0.0231	0.0092	1.000	-
17	DFM	0.0366	0.0133	1.588	0.230

High-dimensional and flexible methods dominate the US rankings. The LSTM cuts the ARMA RMSFE nearly in half with a relative score of 0.502. It is followed closely by the MLP (0.592), the reduced BVAR (0.633), and Ridge regression (0.660). These results match the US findings in Hopp (2022), where LSTM and BVAR models also outperform the field. While these performance gaps are economically large, they are not statistically decisive due to the short sample length.

³See `data/evaluation_results_us.csv`, `turkey_data/evaluation_results_tr.csv`, and `data/evaluation_results_us_improved.csv`.

We must note a caveat for the bottom entry: the US DFM ranks last at the **m3** vintage. Section 5.2 shows that this poor performance is entirely concentrated in the COVID-19 quarters. Because our setup keeps parameters fixed at training-window estimates on the monthly Category-3 panel, this failure does not mean factor models fail in general. In fact, the validation-selected Turkish DFM performs much better. It stays mid-table at **m3** and becomes highly competitive at post-release horizons.

Table 6: Türkiye: 17-model ranking on the full panel (2018-Q1 to 2025-Q4) at the **m3** vintage. DM p -values are two-sided. For models ranking below ARMA, a small p -value indicates significantly worse predictive accuracy than the benchmark.

Rank	Model	RMSFE	MAE	Rel. RMSFE	DM p -value
1	Lasso	0.0190	0.0123	0.503	0.134
2	ElasticNet	0.0194	0.0125	0.515	0.132
3	BVAR	0.0205	0.0158	0.544	0.199
4	OLS	0.0259	0.0168	0.688	0.316
5	VAR	0.0265	0.0176	0.703	0.338
6	MIDAS	0.0265	0.0151	0.705	0.135
7	Ridge	0.0266	0.0160	0.705	0.147
8	DFM	0.0276	0.0180	0.732	0.358
9	MLP	0.0297	0.0176	0.789	0.151
10	Gradient Boosting	0.0316	0.0183	0.840	0.194
11	XGBoost	0.0316	0.0179	0.840	0.170
12	Random Forest	0.0329	0.0177	0.873	0.199
13	DeepVAR	0.0348	0.0172	0.923	0.228
14	ARMA	0.0377	0.0191	1.000	-
15	Decision Tree	0.0391	0.0240	1.037	0.723
16	LSTM	0.0428	0.0192	1.137	0.444
17	MIDAS-ML	0.0530	0.0362	1.407	0.132

The model rankings for Türkiye are nearly inverted. Penalized shrinkage methods and reduced-dimensional econometric models take the top spots. Lasso leads with a relative RMSFE of 0.503, followed by ElasticNet (0.515) and BVAR (0.544). Simple OLS and VAR models also round out the top five. Conversely, neural networks perform poorly in this environment. The MLP sits in the middle of the table, while the LSTM drops to 16th place, failing to beat the simple ARMA benchmark. The last-place finish for MIDAS-ML requires a similar caveat. It uses a constrained, fixed-penalty fallback due to tuning issues (Section 4.4). When properly cross-validated, its US counterpart ranks sixth.

The reduced-dimensional BVAR is the only model that lands in the top tier for both countries. Beyond the top few models, the mid-table rankings are effectively a tie. In the US, ranks 9 through 15 differ by less than 0.001 in absolute RMSFE. They are not statistically distinguishable. The Diebold-Mariano p -values support this view. While they favor the top models, they are not strictly decisive. The best models in both countries yield p -values between 0.13 and 0.24 against ARMA. Given our brief 36-quarter window, extreme COVID-19 shocks naturally inflate the variance of our loss differentials. This weak statistical power is expected, so we avoid making strong claims based purely on significance tests.

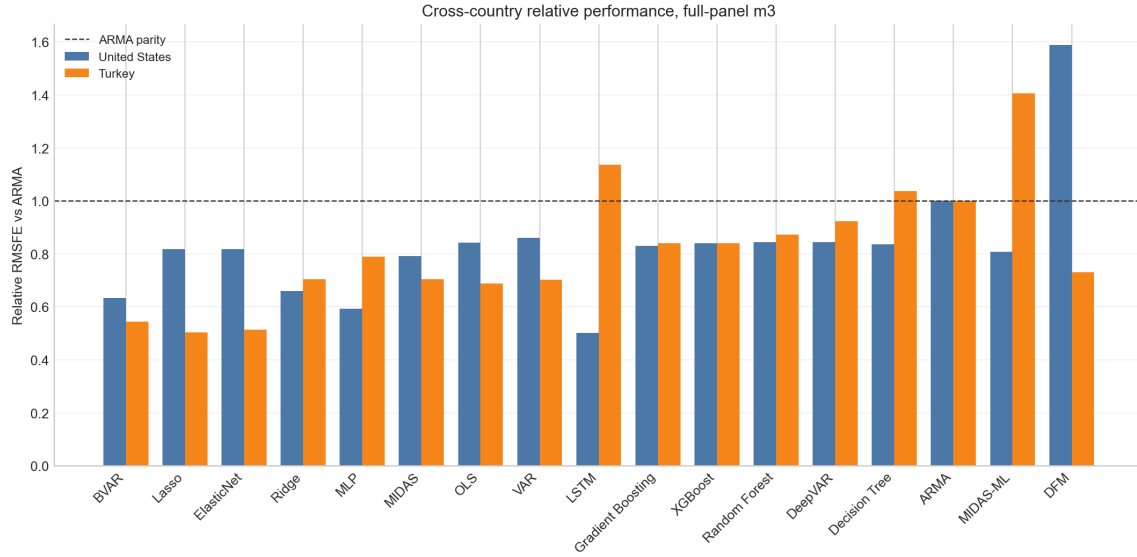


Figure 1: Relative RMSFE against ARMA at the m3 vintage for the US and Türkiye. Scores below one represent an improvement over the univariate benchmark.

Figure 1 visualizes this cross-country comparison on a common scale. The stark contrast in the leading model types is the core result of this paper. US data favors neural and high-dimensional methods, while Turkish data favors shrinkage designs.

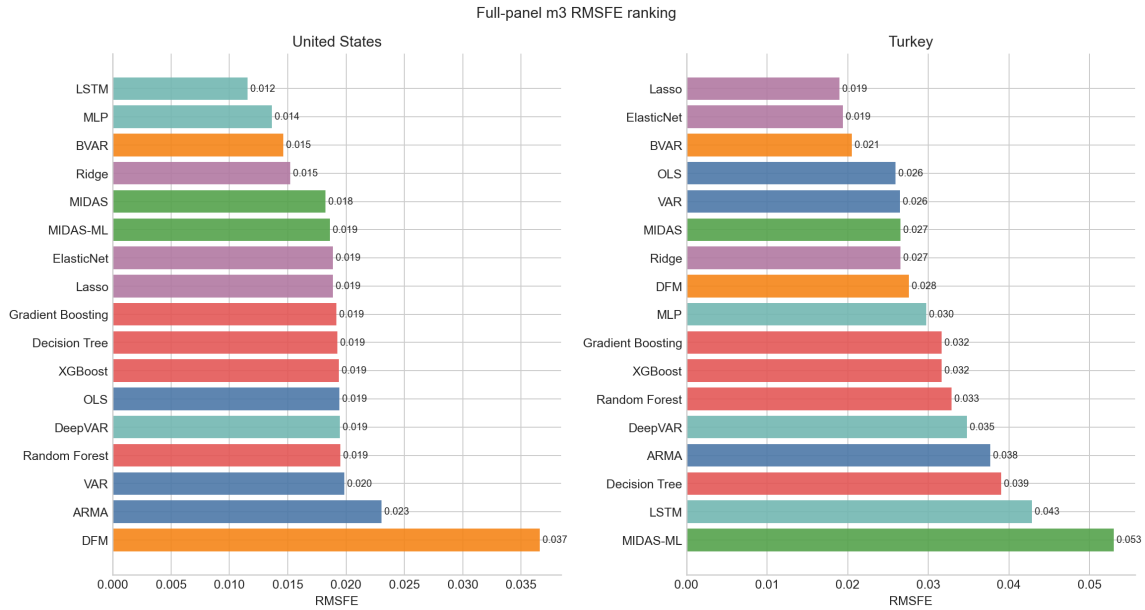


Figure 2: Full-panel RMSFE rankings by country at the m3 vintage.

Figure 2 plots these rankings using absolute RMSFE values. This highlights the clear difference in baseline forecast error scales. Turkish nowcast errors are roughly 1.6 times larger than US errors across the entire distribution, ranging from 1.4 to 1.9 times larger rank by rank.

Table 7: Family-level performance on the full panel at the m3 vintage. Values represent the average relative RMSFE against ARMA across all family members, with the top performer listed in parentheses.

United States			Türkiye		
Family	Avg. rel.	Best	Family	Avg. rel.	Best
Neural	0.646	(LSTM)	Penalized linear	0.574	(Lasso)
Penalized linear	0.766	(Ridge)	Bayesian/factor	0.638	(BVAR)
Mixed-frequency	0.799	(MIDAS)	Classical	0.695	(OLS)
Tree-based ML	0.838	(GB)	Tree-based ML	0.897	(GB)
Classical	0.852	(OLS)	Neural	0.950	(MLP)
Benchmark	1.000	(ARMA)	Benchmark	1.000	(ARMA)
Bayesian/factor	1.111	(BVAR)	Mixed-frequency	1.056	(MIDAS)

Table 7 illustrates this clean contrast at the model-family level. In the US, the neural family achieves the best average relative RMSFE at 0.646. In Türkiye, the penalized linear family leads at 0.574, while neural models fall to second-to-last at 0.950.

However, two family averages require careful interpretation. Both average a single strong model with a constrained counterpart. For instance, the poor US Bayesian/factor average (1.111) mixes a highly successful BVAR (0.633) with a pandemic-sensitive DFM (1.588). Similarly, the weak Turkish mixed-frequency score (1.056) combines a strong standard MIDAS model (0.705) with the restricted MIDAS-ML fallback (1.407). Looking at the individual "Best" column provides a more accurate summary. Both BVAR and MIDAS are robust across both countries. Their high family averages stem from specific implementation bottlenecks rather than flaws in the underlying methodology.

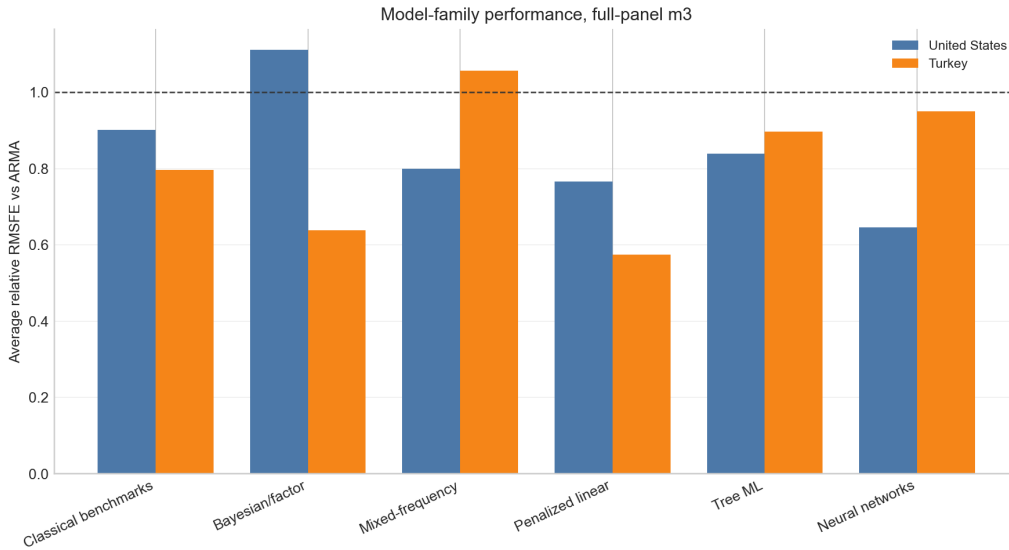


Figure 3: Average relative RMSFE by model family against ARMA at the m3 vintage.

Figure 3 charts these family averages. This layout highlights how completely the model rankings change between the two data environments.

5.2 Period-Specific Performance and the COVID Stress Test

A few volatile quarters can easily distort full-sample RMSFE metrics. To address this, we isolate distinct sub-periods. Table 8 shows the top five models for each sub-period using absolute RMSFE values. Table 9 shows how the full-sample leaders perform across these separate windows.

Table 8: Top five models and their absolute RMSFE values across sub-periods at the `m3` vintage.

Period	Top five models
<i>United States</i>	
Pre-COVID	GB (0.0026), OLS (0.0027), VAR (0.0027), DeepVAR (0.0027), MIDAS (0.0027)
COVID	LSTM (0.0235), BVAR (0.0260), MLP (0.0288), Ridge (0.0326), MIDAS (0.0399)
Post-COVID	DeepVAR (0.0043), XGBoost (0.0044), MIDAS-ML (0.0045), MIDAS (0.0045), OLS (0.0045)
Full	LSTM (0.0116), MLP (0.0137), BVAR (0.0146), Ridge (0.0152), MIDAS (0.0182)
<i>Türkiye</i>	
Pre-crisis	MLP (0.0085), ElasticNet (0.0094), DFM (0.0095), Ridge (0.0098), Lasso (0.0098)
COVID	BVAR (0.0253), Lasso (0.0296), ElasticNet (0.0311), OLS (0.0430), VAR (0.0431)
Post-COVID	ARMA (0.0116), MIDAS (0.0117), DeepVAR (0.0119), LSTM (0.0120), RF (0.0121)
Full	Lasso (0.0190), ElasticNet (0.0194), BVAR (0.0205), OLS (0.0259), VAR (0.0265)

The pandemic serves as our primary stress test, and it significantly reshuffles the model rankings. The full-sample leaders (LSTM, MLP, BVAR, and Ridge) completely disappear from the top five slots during calm periods. Instead, tree-based methods, classical models, and DeepVAR take over. During stable economic times, performance gaps between models shrink drastically. In the pre-COVID US sample, the top five models are separated by a mere 10^{-4} in RMSFE. Simple benchmarks like OLS and VAR rank near the top.

DeepVAR’s strong showing in calm periods requires a caution based on its vintage insensitivity (Section 4.7). Its predictions are mostly flat, which naturally minimizes errors when economic volatility is low. This reflects a structural artifact rather than genuine forecasting skill.

The COVID-19 window reveals our primary cross-country pattern clearly. In the US, the full-sample leaders (LSTM, BVAR, and MLP) handle the shock best. In Türkiye, the pandemic rankings are led by BVAR, Lasso, and ElasticNet, emphasizing shrinkage and low-dimensional constraints. Because pandemic-era errors are an order of magnitude larger than normal quarters, COVID performance dictates the full-sample results. This justifies isolating these sub-periods rather than relying on a single pooled table. Appendix A, Figure 9 confirms that these sub-period rankings remain robust.

Table 9: Relative RMSFE against ARMA across sub-periods at the `m3` vintage for the leading models. For Türkiye, we include the top four models plus the validation-selected DFM. Values below one show improvement over ARMA. Values above one mean the model underperformed the baseline. The sub-period columns highlight that the gains are heavily concentrated within the pandemic window.

Model	Pre-COVID / pre-crisis	COVID	Post-COVID	Full
<i>United States</i>				
LSTM	1.250	0.460	1.024	0.502
MLP	1.165	0.565	1.013	0.592
BVAR	1.972	0.510	2.413	0.633
Ridge	1.122	0.638	1.039	0.660
MIDAS	0.986	0.783	0.943	0.791
<i>Türkiye</i>				
Lasso	0.527	0.390	1.447	0.503
ElasticNet	0.504	0.410	1.453	0.515
BVAR	0.636	0.333	1.899	0.544
OLS	0.900	0.565	1.688	0.688
DFM	0.510	0.598	2.050	0.732

Table 9 illustrates exactly how concentrated these predictive gains are. The massive full-sample improvements over ARMA are almost exclusively tied to the pandemic shock. In the US, BVAR’s relative RMSFE stands at 0.51 during COVID, but shoots up to 1.97 pre-COVID and 2.41 post-COVID. Similarly, the LSTM relative error shifts from 0.46 during COVID to 1.25 and 1.02 in the quiet periods. Outside of crises, the top US models generally fail to beat the simple ARMA baseline.

Our alternative `non_covid` evaluation panel confirms this pattern. When we drop the entire 2020-Q2 to 2021-Q4 window, the top model barely reaches a relative RMSFE of 0.94. Under those same calm conditions, BVAR spikes to 1.80 and the LSTM rises to 1.03.

Türkiye follows the exact same pattern, though on a slightly different timeline. Its top performers (Lasso, ElasticNet, BVAR) and the DFM easily beat ARMA during the 2018 currency crisis and the pandemic. However, they underperform significantly from 2022 to 2025, posting relative errors of 1.45, 1.45, 1.90, and 2.05. As a result, ARMA wins the post-COVID Turkish sample outright.

The broader takeaway is clear. Sophisticated models earn their edge during volatile periods when univariate benchmarks fail. During stable periods, model differences vanish and ARMA becomes highly competitive. In fact, it is the most accurate model for post-COVID Türkiye. The Turkish pre-crisis window seems like an exception, but it actually confirms this rule. That period includes the 2018 currency crisis, where complex models beat ARMA easily, leaving it in 15th place. Pooled rankings are a reflection of crisis performance, not an all-weather standard.

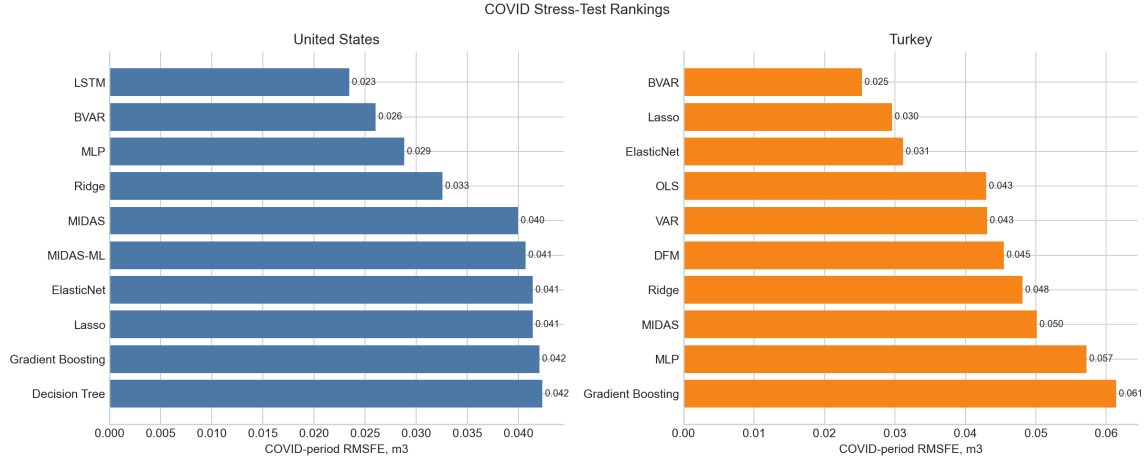


Figure 4: COVID-period stress test at the m3 vintage. The chart tracks absolute COVID-panel RMSFE alongside its ratio to calm-period performance for the top models in each country.

Figure 4 provides a visual summary of this stress test. It contrasts absolute pandemic errors against baseline calm periods, illustrating how dramatically errors spike and how unevenly the shock impacts each architecture.

As a final check on robustness, we count how often each model lands in the top three across all sub-periods. The most reliable US models are the LSTM, BVAR, and MLP. For Türkiye, the consistent leaders are ElasticNet (top three in three out of four sub-periods), Lasso, BVAR, and the DFM (see Appendix A, Figure 8). These stable groups line up perfectly with our full-sample leaders. This proves that our central cross-country contrast is shaped entirely by high-volatility panels. In calm periods, performance differences flatten out, and the structural divide between the two countries disappears.

5.3 Within-Quarter Information Gains (m1→m3)

Nowcast accuracy should naturally improve as a quarter unfolds and new data points arrive. This subsection evaluates that improvement by tracking shifts from our earliest data vintage (m1) to our final within-quarter update (m3).

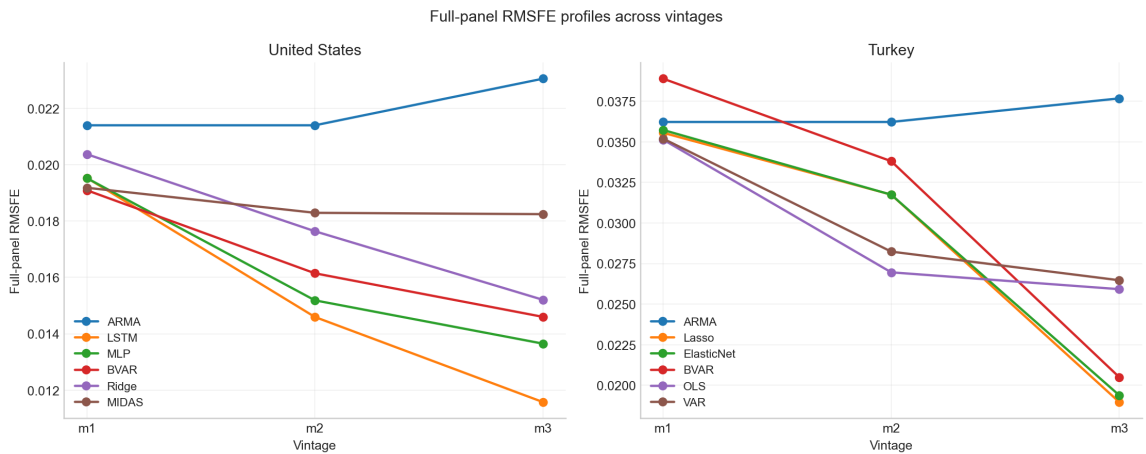


Figure 5: Full-panel RMSFE tracking across m1, m2, and m3 vintages for ARMA and the leading models. Accuracy generally climbs as data accumulates, but the rate of improvement varies across model types.

Figure 5 shows that accuracy improves over time, but the gains are highly uneven. At the earliest vintage (**m1**), performance differences are minimal. In Türkiye, all multivariate models cluster tightly between 0.034 and 0.039 RMSFE. The top performers only separate themselves as monthly indicators pour in. This means information timing is critical. A model’s rank depends heavily on how much data has been observed.

Table 10: Full-panel within-quarter information gains for headline models. The table reports RMSFE at the **m1** and **m3** vintages alongside the percentage improvement. Note that for Türkiye, the standard VAR and MIDAS models also improve by roughly 25%.

United States				Türkiye			
Model	m1	m3	Gain %	Model	m1	m3	Gain %
LSTM	0.0195	0.0116	40.7	BVAR	0.0389	0.0205	47.3
MLP	0.0195	0.0137	30.1	Lasso	0.0356	0.0190	46.7
Ridge	0.0204	0.0152	25.4	ElasticNet	0.0357	0.0194	45.7
BVAR	0.0191	0.0146	23.5	Ridge	0.0361	0.0266	26.3
Lasso	0.0199	0.0189	5.0	OLS	0.0351	0.0259	26.2
MIDAS	0.0192	0.0182	4.9	DFM	0.0341	0.0276	19.3

Table 10 quantifies these dynamic improvements. In the US, the sharpest drops in error from **m1** to **m3** occur in the LSTM (40.7%), MLP (30.1%), Ridge (25.4%), and BVAR (23.5%). In Türkiye, the biggest gainers are BVAR (47.3%), Lasso (46.7%), and ElasticNet (45.7%), while Ridge, OLS, VAR, and MIDAS improve by around 25%, and the DFM gains 19.3%.

Crucially, the models that benefit most from incoming information are the top performers in each country. This indicates that their core advantage comes from converting new data into accurate revisions, rather than starting the quarter with superior accuracy.

The US DFM serves as a clear warning here. Its error actually increases as the quarter progresses, with absolute RMSFE rising from 0.0161 to 0.0366. Because it lacks a pandemic dummy and relies on fixed training parameters, it misinterprets highly unusual monthly data as the quarter develops. Incoming data only helps models structured to ingest it.

5.4 Post-Release Robustness Horizons

Post-release horizons (**post1** for both nations, plus **post2** for Türkiye) evaluate a distinct scenario. How much does accuracy improve when we observe post-quarter data while the target GDP release remains unpublished? This question is vital for Turkish policymakers because official quarterly GDP figures face long publication lags 2 months.

Figure 6 below, shows that the US **post1** ranking mirrors our **m3** results closely, reinforcing the baseline conclusions. There is only one notable exception: BVAR drops slightly below Ridge regression, yielding errors of 0.0154 compared to Ridge’s 0.0147. BVAR is the only top model that fails to improve with extra post-quarter observations. Its relative RMSFE edges up from 0.633 to 0.667. This fits its design as an instrument tailored for within-quarter crisis detection.

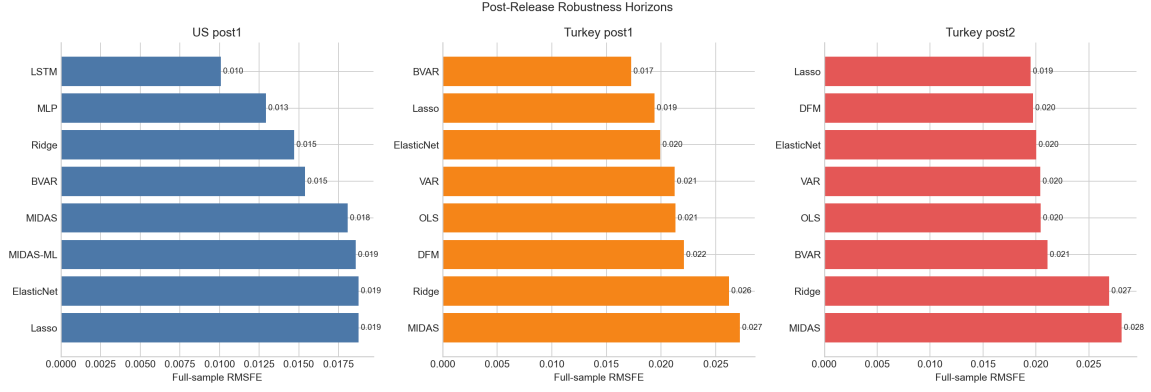


Figure 6: Post-release robustness rankings across full panels for the US (**post1**) and Türkiye (**post1** and **post2**).

In contrast, the post-release horizons provide massive informational gains for Türkiye. At the **post1** horizon, BVAR climbs to first place with a relative RMSFE of 0.459. By the **post2** horizon, the validation-selected DFM surges to second place (0.0197), trailing only Lasso (0.0195). The dynamic factor model goes from mediocre within the quarter to highly accurate once hard, post-quarter data arrives. This performance is expected when fresh indicators flow into a Kalman filter with fixed parameters. Given Türkiye’s publication delays, this is a highly practical solution for real-time tracking, though it does not change our core within-quarter (**m1** to **m3**) conclusions. Appendix A, Table 11 details the full post-release rankings.

5.5 Forecast Combinations: A US Robustness Extension

As an extension, we evaluate simple forecast combinations for the US by averaging predictions from selected base models. Because we designed these combinations after reviewing baseline model performance, we treat them as a secondary robustness check separate from the main benchmark.

Our main finding is that an *equal-weight* combination of BVAR, MIDAS, and the DFM achieves an impressive full-sample **m3** RMSFE of 0.0074. This gives a relative score of 0.319, outperforming every individual model. This success stems from error cancellation during volatile quarters. In 2020-Q2, the DFM vastly overestimates the economic contraction, while BVAR and MIDAS underestimate it. Averaging these predictions cancels out the dominant errors. Because pandemic quarters account for most of our full-sample squared loss, smoothing these massive errors drops the overall RMSFE.

This result does not depend on a single quarter. The combination keeps its top spot even when we exclude 2020-Q2 alone, yielding an RMSFE of 0.0065. However, if we drop the *entire* pandemic window (2020-Q2 to 2021-Q4), its relative RMSFE rises to 1.18, slightly underperforming the baseline. This confirms that the combination’s edge is concentrated in turbulent regimes.

Machine learning ensembles and trimmed means also outperform their individual parts. This pattern demonstrates a classic forecasting rule: blending distinct, complementary methodologies often yields better results than trying to pick a single winning model. Full details are available in Appendix A, Table 12 and Figure 7.

6 Discussion

6.1 Why Do the Rankings Differ Across Countries?

The central empirical finding of this paper is that an identical 17-model lineup produces nearly opposite rankings in the US and Türkiye. This divergence is driven by three primary differences in their data environments.

First, *sample length*. The US models train on roughly 200 quarterly observations. For sequence models, this translates to several times more monthly observations. In contrast, the Turkish models are heavily constrained by data availability. While some data goes back to 1995, the broader indicator panel only becomes reliable around 2005. Many Turkish series begin in the mid-2000s, leaving earlier rows largely filled with mean-imputed values that carry little indicator signal. The effective informative sample is therefore on the order of 50 quarters, which is several times shorter than the US history. Flexible, high-capacity methods (*the LSTM and MLP in particular*) exploit the long US history without overfitting. However, on the Turkish panel, these same architectures struggle to learn from a small sample supplemented with constant dummy variables. Cross-validation results in Sections 4.5 and 4.6 show this constraint directly. For penalized regressions and most tree-based models, cross-validation selects substantially heavier regularization for Türkiye than for the United States. Meanwhile, the neural architectures are given identical settings in both countries, and the random forest results are mixed.

Second, *data richness*. The US panel offers genuinely high-dimensional information for models that can absorb it. It includes 53 indicators distilled from a 296-variable pool, alongside weekly financial series. The Turkish panel contains only 22 usable indicators, which is less than half the size of the US pool. Furthermore, many of these Turkish series have short histories. This leaves less cross-sectional signal for high-dimensional methods to extract, compressing their advantage over simpler, parsimonious specifications.

Third, *volatility and structural instability*. Turkish GDP growth over the evaluation window is roughly twice as volatile as US growth. The standard deviation of quarterly GDP growth is about 0.035 for Türkiye against 0.019 for the United States, a ratio near 1.8. Additionally, the Turkish sample contains major structural shocks: the 2018 currency crisis, the COVID pandemic, and the post-2021 inflation surge. In such a noisy environment, estimation variance is costly and shrinkage pays. The models that survive are those that pull aggressively toward parsimonious benchmarks like Lasso, ElasticNet, and the Minnesota-prior BVAR. Flexible models simply chase noise.

These mechanisms are consistent with the comparative literature. Our US result (LSTM and BVAR at the top) reproduces the findings of [Hopp \(2022\)](#) on an extended sample. Meanwhile, our Türkiye result aligns with [Akepanidtaworn and Akepanidtaworn \(2025\)](#), who find that traditional methods and *simple* machine learning (lasso, elastic net) dominate complex algorithms when samples are short. This divergence is highly informative. As shown in the worldwide survey by [Cascaldi-Garcia et al. \(2023\)](#), nowcasting practice must adapt to local data environments rather than following a single template. Viewed through this lens, the two countries represent opposite ends of the signal-to-noise spectrum. This spectrum dictates the trade-off between econometrics and machine learning. Finally, this cross-country contrast is itself regime-dependent.

As Section 5.2 shows, the gap between the winning model classes is widest under economic stress. During calm quarters, performance in both countries compresses toward simple ARMA benchmarks. Thus, structural factors explain which classes pull ahead *when there is a shock to forecast*, rather than establishing a permanent, all-weather ranking.

6.2 Model-Specific Lessons

Beyond broad model families, four specific model-level findings deserve emphasis.

BVAR is the most portable model across countries, though its skill is strongly concentrated in crisis regimes. The reduced-dimensional BVAR is the only model to rank in the top three for both countries' full-sample `m3` rankings. Disciplined multivariate shrinkage with a default Minnesota prior travels across data environments better than any other design choice in the lineup. However, this cross-country robustness sits alongside a sharp cross-regime concentration. BVAR leads the Türkiye COVID and `post1` panels, posting large within-quarter information gains. Outside of crises, it is one of the weakest leaders. Its relative RMSFE rises from 0.51 during COVID to 1.97 and 2.41 in the US pre- and post-COVID panels. Similarly, it rises from 0.33 to 1.90 in the Türkiye post-COVID panel (Table 9). Its high MAE relative to its RMSFE rank (Table 5) shows the same pattern from a different angle. The model controls the large squared errors of the crisis tail at the cost of larger typical errors in calm quarters. BVAR is therefore best viewed as a portable crisis hedge rather than an a uniformly dominant benchmark.

The neural models are the sharpest single expression of the cross-country reversal. The LSTM is the best US model at `m3` (relative RMSFE 0.50), yet it ranks sixteenth in Türkiye, falling behind a simple ARMA model (1.14). The MLP follows the exact same pattern. Their high parameter capacity is rewarded on long, data-rich US series. However, it becomes a liability on the short, noisy Turkish panel, where cross-validation selects the heaviest available regularization. This single pair of models accounts for much of the headline contrast. It clearly illustrates that flexible architectures are an asset only when the data environment can support them.

The DFM's weakness is an implementation result, not a verdict on factor models. The US DFM ranks last at `m3` (relative RMSFE 1.59), but this failure is entirely concentrated in 2020. Because it was estimated on a fixed training window without COVID dummies, it could not absorb the pandemic shock. Its error actually grew as within-quarter data arrived (Section 5.3). Conversely, the validation-selected Turkish DFM performed decently within the quarter and rose to second place at `post2` once post-quarter hard data became available. Read together, these results show that factor models are highly sensitive to specification and pandemic breaks, rather than being uncompetitive in principle.

MIDAS-ML should be read as a constrained build, not as an indictment of the method. Its last-place Turkish result reflects a fixed-penalty fallback used when rolling cross-validation failed to complete (Section 4.4). In contrast, its cross-validated US counterpart ranks sixth. We therefore draw no broad method-class conclusion from the Turkish MIDAS-ML numbers. More generally, the period tables caution against viewing any model's full-sample rank as a fixed trait. Model-level lessons remain highly conditional on the volatility regime.

6.3 Implications for Practice

For practitioners, three main implications follow. First, model choice must be conditioned on the local data environment. Directly importing a US-validated nowcasting architecture into an emerging market framework is unsupported by empirical evidence. Second, within-quarter information timing matters as much as model choice. The performance gap between the best and worst models at $m1$ is small in both countries. Early in the quarter, investing in timely data acquisition yields higher returns than investing in model sophistication. Third, for institutions nowcasting Turkish GDP specifically, long publication lags make post-release horizons operationally vital. In these horizons, Lasso remains the most consistent performer, while the validation-selected DFM and BVAR show notable improvements in accuracy (consistent with the DFM tradition in Turkish literature; [Modugno et al., 2016](#); [Soybilgen and Yazgan, 2017](#)).

7 Conclusion

This paper presents a pseudo-real-time comparison of 17 GDP nowcasting models for the United States and Türkiye. The model lineup remains fixed, while the underlying data environment varies. To simulate actual data availability, we use publication-lag and ragged-edge masking across three standardized within-quarter vintages. We evaluate all models against final revised GDP values. The test period covers 36 quarters for the US (2017–2025) and 32 quarters for Türkiye (2018–2025). We report COVID and non-COVID periods separately to isolate the extreme volatility of the pandemic. Finally, we include post-release horizons as a robustness check.

The findings are fivefold. First, performance does not translate across different data environments, especially during periods of economic stress. In the US, flexible high-dimensional methods like LSTM, MLP, and reduced BVAR perform best. In contrast, shrinkage and reduced-dimensional econometric models like Lasso, ElasticNet, and BVAR lead in Türkiye. Notably, the LSTM underperforms the simple univariate benchmark when applied to the Turkish data environment. Second, machine learning models are not universally superior. Instead, their success depends heavily on data richness, sample length, and macroeconomic volatility. This conditionality helps explain the conflicting findings found in previous comparative literature. Third, the chosen evaluation period heavily influences the results. Model rankings during the COVID-19 pandemic differ significantly from calm periods, yet pandemic quarters dominate the overall statistics. Because of this, single-number summaries are misleading. Top-performing models build most of their advantage during high-stress periods. During calm quarters, the univariate benchmark remains highly competitive. In Türkiye’s post-COVID quarters, the univariate benchmark actually performs best. Fourth, within-quarter information timing is substantive: accuracy improves by up to 40-47% from the first to the third month of the quarter for the best models, which are precisely the models that convert incoming data into revisions most effectively. Fifth, the reduced-dimensional BVAR is the most robust single model across countries, periods, and horizons, and in the United States an equal-weight combination of econometric models (BVAR, MIDAS, DFM) outperforms every individual model as a robustness extension.

Limitations

The limitations are stated explicitly. The exercise is pseudo-real-time: publication lags and ragged edges are simulated, but the underlying series are final revised data, so revision uncertainty is not measured and no claim is made about operational real-time accuracy. The three within-quarter vintages are coarse information sets, not the full release-by-release data flow, and no news decomposition is performed. Several implementations are constrained by computational feasibility: the BVARs are reduced-dimensional, the Turkish DFM is a validation-selected Category-2 specification rather than a full sparse-panel factor model, and the Turkish MIDAS-ML uses a fixed penalty after cross-validation failed to complete. The COVID dummies reflect ex-post knowledge of the pandemic quarters. Evaluation panels are short, so Diebold-Mariano comparisons are directional rather than decisive, and the Turkish MLP exhibits early-vintage instability that we report as a caveat rather than a result.

Future Research

Three extensions follow naturally. First, a true historical-vintage evaluation using archival databases such as ALFRED would measure the revision channel this study deliberately set aside. Second, release-level news decomposition in the DFM/BVAR framework would convert the vintage-level results into indicator-level attribution, as in [Modugno et al. \(2016\)](#). Third, and most promising for Türkiye, alternative high-frequency data (card transactions, payments, and other timely indicators, in the spirit of [Barlas et al., 2024](#)) could relax the early-vintage information constraint that compresses model performance at $m1$, where all models are weakest and the policy value of a reliable nowcast is greatest.

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A Additional Results

Table 11: Post-release robustness rankings, full panels (top eight models by relative RMSFE versus ARMA).

US <i>post1</i>		Türkiye <i>post1</i>		Türkiye <i>post2</i>	
Model	Rel.	Model	Rel.	Model	Rel.
LSTM	0.437	BVAR	0.459	Lasso	0.517
MLP	0.561	Lasso	0.515	DFM	0.524
Ridge	0.638	ElasticNet	0.528	ElasticNet	0.532
BVAR	0.667	VAR	0.564	VAR	0.541
MIDAS	0.784	OLS	0.566	OLS	0.543
MIDAS-ML	0.807	DFM	0.587	BVAR	0.560
ElasticNet	0.814	Ridge	0.696	Ridge	0.715
Lasso	0.814	MIDAS	0.723	MIDAS	0.746

Table 12: US forecast-combination robustness layer, full panel, *m3* vintage (selected entries). Combinations are equal-weight averages of the indicated base-model predictions and are reported separately from the 17-model benchmark because they were constructed after observing base-model behaviour.

Model / combination	RMSFE	Rel. RMSFE
Combination: BVAR + MIDAS + DFM	0.0074	0.319
LSTM	0.0116	0.502
MLP	0.0137	0.592
Combination: LSTM + BVAR + MIDAS	0.0139	0.602
BVAR	0.0146	0.633
Ridge	0.0152	0.660
Combination: ML (LSTM, RF, GB, XGB, MLP)	0.0165	0.717
Combination: all models, trimmed mean	0.0173	0.752
Combination: penalized linear	0.0176	0.764
MIDAS	0.0182	0.791

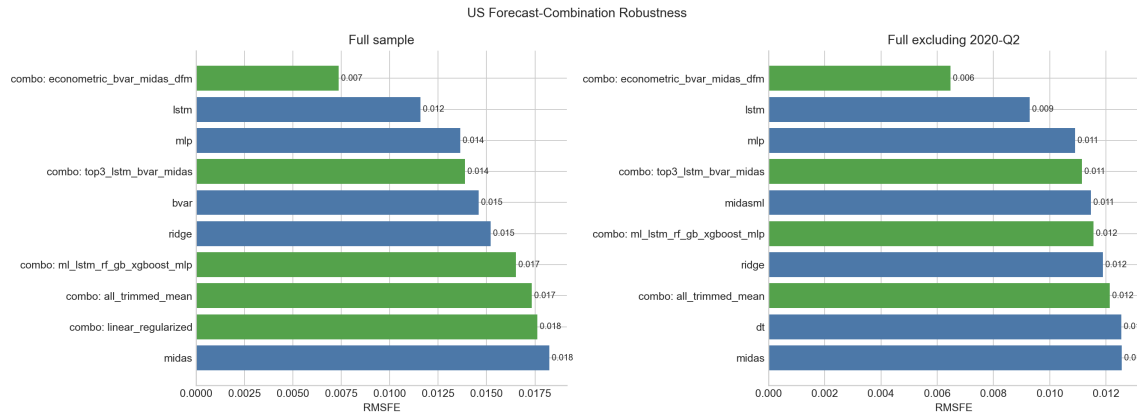


Figure 7: US forecast-combination robustness: full-sample and excluding-2020-Q2 rankings including combination forecasts.

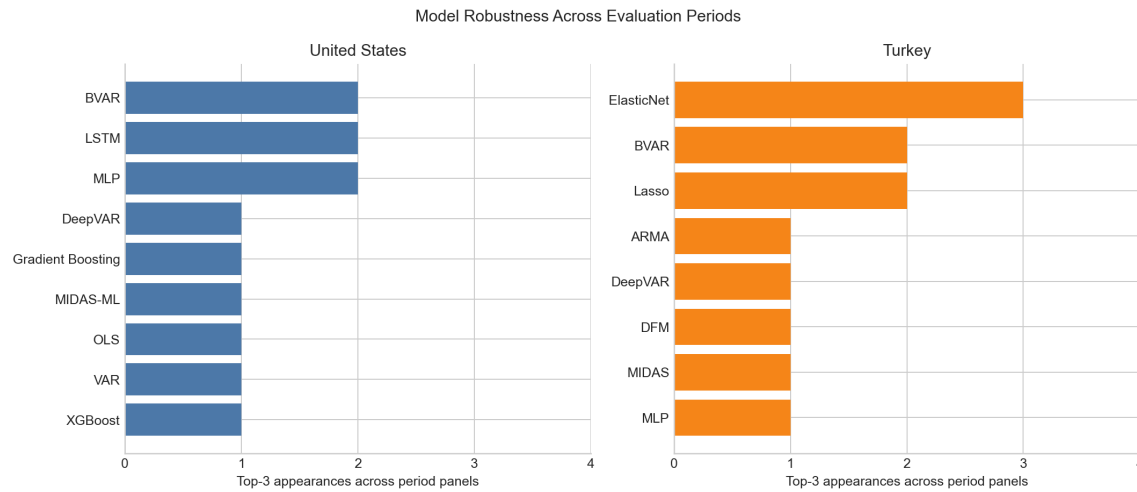


Figure 8: Robustness across regimes: number of top-three appearances across the period panels (m3 vintage), by country.

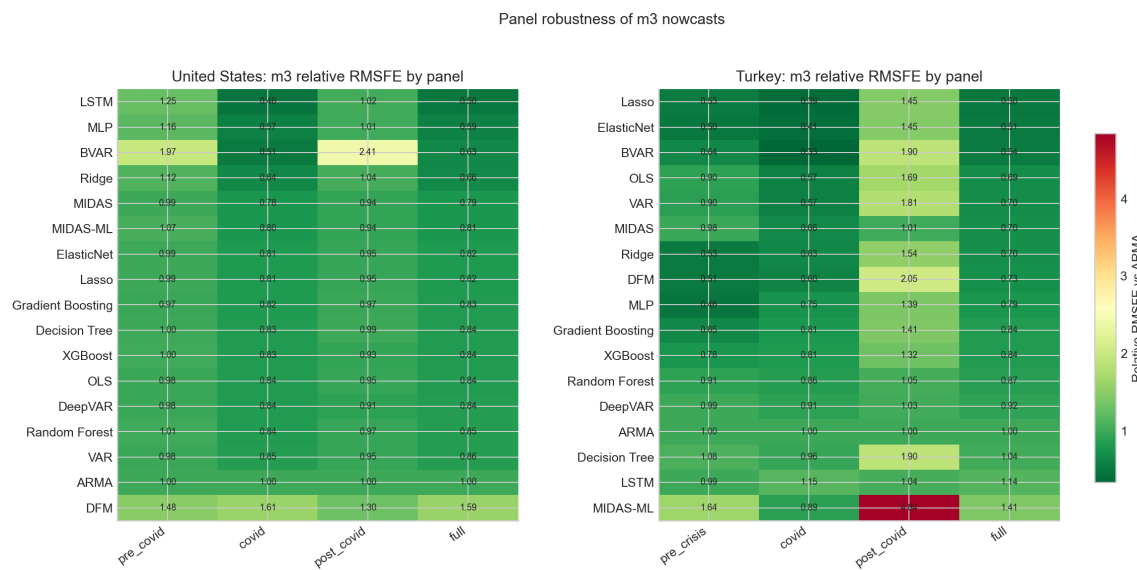


Figure 9: Panel robustness of m3 relative RMSFE: heatmaps across models and evaluation panels, by country.

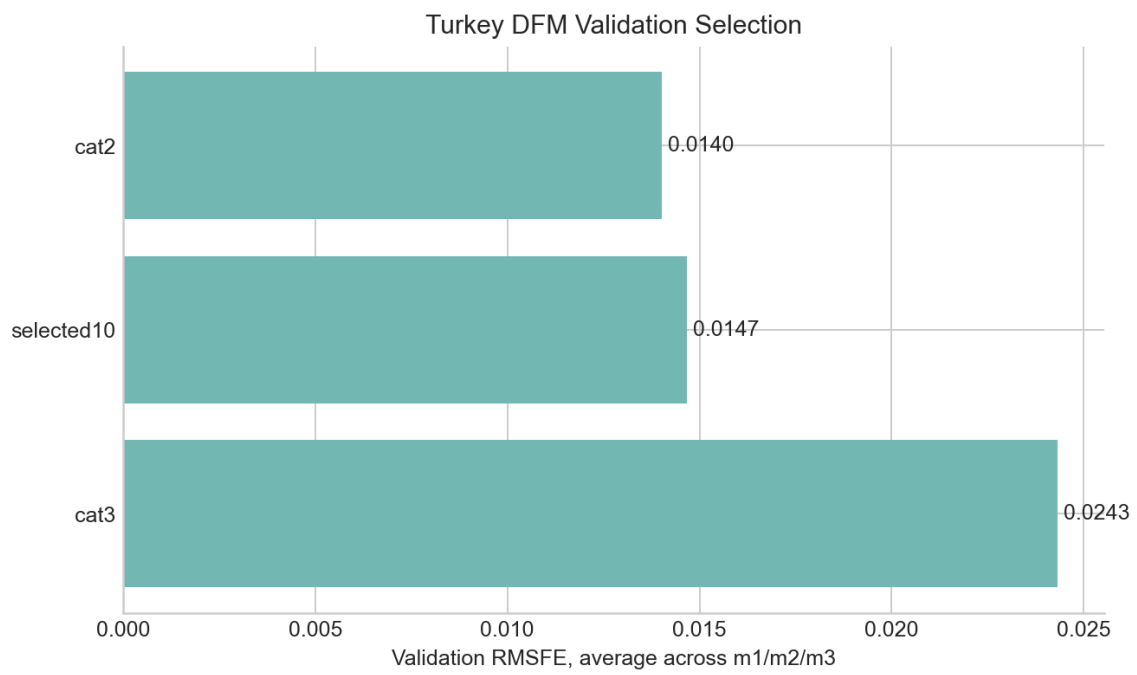


Figure 10: Türkiye DFM validation-based specification selection: candidate predictor panels compared on 2012-2017 validation RMSFE across the $m1/m2/m3$ vintages. The Category-2 panel was selected and retrained through 2017 before test evaluation.

Table 13: Selected cross-validated hyperparameters by country. All tuning uses time-ordered cross-validation on the training-plus-validation window, with scaling applied inside the cross-validation pipeline; selected values are frozen for the rolling test loop. All models share the same validation and test windows; the training window also coincides except for the mixed-frequency models (MIDAS, MIDAS-ML), whose start dates (US 1967-Q1, Türkiye 2002-Q1) are set by the availability of the weekly indicators they require.

Model	United States	Türkiye
ARMA	ARMA(2,0), AIC-selected	ARMA(2,0), AIC-selected
VAR	lag 1 (AIC)	lag 1 (AIC)
BVAR	Minnesota prior, $\lambda = 0.1$ (fixed default); Lasso-80 predictor set	Minnesota prior, $\lambda = 0.1$ (fixed default); Category-2 + target
DFM	monthly Cat-3 + target; block structure; EM; estimated once	validation-selected Cat-2 + target; EM; estimated once
MIDAS	nealmon weights; $J=4$ monthly, $J=13$ weekly; inverse in-sample-RMSE combination	same lag structure; 2 weekly indicators; distance-from-maximum combination
MIDAS-ML	Legendre degree 3, $\gamma=0.25$, CV-selected λ	same, fixed $\lambda = 0.01$ (fallback)
Lasso	$\hat{\alpha} = 2.5 \times 10^{-5}$	$\hat{\alpha} = 1.6 \times 10^{-3}$
Ridge	$\hat{\alpha} = 14.6$	$\hat{\alpha} = 259.5$
ElasticNet	$\hat{\alpha} = 2.5 \times 10^{-5}$, $\hat{\rho} = 1.00$ (= Lasso)	$\hat{\alpha} = 1.5 \times 10^{-2}$, $\hat{\rho} = 0.10$ (Ridge-leaning)
Decision Tree	depth unbounded, min. leaf 5	depth 3, min. leaf 5
Random Forest	200 trees, depth 8, min. leaf 3	200 trees, unbounded depth, min. leaf 5
Gradient Boosting	rate 0.01, 500 trees, depth 3	rate 0.05, 100 trees, depth 3, subsample 0.8
XGBoost	depth 5, $\eta=0.1$, 500 trees, $\lambda_{\ell_2}=10$	depth 3, $\eta=0.05$, 100 trees, $\lambda_{\ell_2}=5$
MLP	layers (32,16), ℓ_2 decay 1.0, robust scaling, early stopping; median of 10 networks	same
LSTM	1 layer, 32 units, 6-month context, 10-network ensemble	same
DeepVAR	4 layers, 40 units, 12-month context (fixed)	same